

LINUX INTERVIEW KIT

300+ Most Asked Linux Interview Questions & Short Answers

```
$ whoami
future_linux_engineer

$ ./crack_the_interview.sh --level=expert
[OK] 260 questions loaded ... ready.
```

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2026 Edition

DevOps Interview Kit
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Total core interview questions: **260** — numbered continuously Q1 to Q260. Bonus sections add 100 more rapid-fire and MCQ items plus two revision tables, taking the kit past 300 questions.

How to revise: read the Short Answer aloud in under 20 seconds, then attempt the Follow-up before checking the next question. Finish every sitting with one pass of the Cheat Code table.

SECTION 1

Basic Linux Questions

Foundations every candidate must answer without hesitation

Q1. What is Linux?

BASIC

SHORT ANSWER

Linux is an open-source, Unix-like operating system kernel. Combined with GNU utilities and other software, it forms complete distributions (RHEL, Ubuntu, Debian) used widely on servers, cloud, and embedded systems.

INTERVIEW TIP

Say 'kernel' first, then 'distribution' — interviewers check if you know the difference.

POSSIBLE FOLLOW-UP

What is the difference between Linux and Unix?

Q2. What is the Linux kernel?

BASIC

SHORT ANSWER

The kernel is the core of the OS. It manages CPU scheduling, memory, devices, filesystems, and system calls, acting as the bridge between hardware and user-space applications.

INTERVIEW TIP

Mention 'monolithic kernel with loadable modules'.

POSSIBLE FOLLOW-UP

What are kernel modules and how do you load one?

Q3. What is a Linux distribution?

BASIC

SHORT ANSWER

A distribution packages the Linux kernel with GNU tools, a package manager, libraries, and utilities. Examples: RHEL, CentOS Stream, Ubuntu, Debian, SUSE, Amazon Linux.

INTERVIEW TIP

Name the distro used in your current project and its version.

POSSIBLE FOLLOW-UP

Which distro have you used in production and why?

Q4. What is the difference between Linux and Unix?

BASIC

SHORT ANSWER

Unix is a proprietary, older OS family (AIX, HP-UX, Solaris); Linux is a free, open-source Unix-like kernel. Linux follows POSIX standards but shares no Unix source code.

INTERVIEW TIP

Keep it short — licensing and origin are the key points.

POSSIBLE FOLLOW-UP

Is macOS Unix-based or Linux-based?

Q5. What is a shell?**BASIC****SHORT ANSWER**

A shell is a command-line interpreter that takes user commands and passes them to the kernel for execution. Common shells: bash, sh, zsh, ksh, fish.

INTERVIEW TIP

Know your default shell: `echo $SHELL`.

POSSIBLE FOLLOW-UP

What is the difference between bash and sh?

Q6. What is the root user?**BASIC****SHORT ANSWER**

root (UID 0) is the superuser with unrestricted access to all commands, files, and system resources. Direct root login is usually disabled; admins use sudo instead.

INTERVIEW TIP

Always add 'we restrict root and use sudo with auditing' — it shows production maturity.

POSSIBLE FOLLOW-UP

How do you disable direct root SSH login?

Q7. What is the Linux file system hierarchy?**BASIC****SHORT ANSWER**

Linux uses a single inverted tree starting at /. Key directories: /etc (configs), /var (logs/variable data), /home (users), /bin & /usr/bin (binaries), /tmp (temporary), /boot (kernel), /proc & /sys (virtual).

INTERVIEW TIP

Quote the FHS (Filesystem Hierarchy Standard) by name.

POSSIBLE FOLLOW-UP

What is stored in /proc?

Q8. What is stored in /etc?**BASIC****SHORT ANSWER**

/etc holds system-wide configuration files — e.g., /etc/passwd, /etc/fstab, /etc/ssh/sshd_config, /etc/hosts. It contains no binaries.

INTERVIEW TIP

Give 2–3 concrete file examples, not just the definition.

POSSIBLE FOLLOW-UP

What is /etc/fstab used for?

Q9. What is /proc?**BASIC****SHORT ANSWER**

/proc is a virtual filesystem exposing kernel and process information as files — e.g., /proc/cpuinfo, /proc/meminfo, /proc/<PID>/ for each process. It exists in memory, not on disk.

INTERVIEW TIP

Mention you use it for live troubleshooting (`cat /proc/meminfo`).

POSSIBLE FOLLOW-UP

What is the difference between /proc and /sys?

Q10. What is an inode?**BASIC****SHORT ANSWER**

An inode is a data structure storing file metadata: permissions, owner, size, timestamps, and pointers to data blocks. It does not store the file name — directories map names to inode numbers.

INTERVIEW TIP

Add: 'a filesystem can run out of inodes even with free space' — a classic production issue.

POSSIBLE FOLLOW-UP

How do you check inode usage?

Q11. What is the difference between a hard link and a soft link?**BASIC****SHORT ANSWER**

A hard link is another name pointing to the same inode; it survives deletion of the original and cannot cross filesystems. A soft (symbolic) link is a separate file pointing to a path; it breaks if the target is deleted and can cross filesystems.

INTERVIEW TIP

Demonstrate: `ln file hard`; `ln -s file soft`.

POSSIBLE FOLLOW-UP

Can you hard-link a directory?

Q12. How do you check the current directory?**BASIC****SHORT ANSWER**

`pwd` prints the absolute path of the current working directory.

INTERVIEW TIP

Mention `$PWD` variable as an alternative.

POSSIBLE FOLLOW-UP

What is the difference between absolute and relative paths?

Q13. How do you list files including hidden files?**BASIC****SHORT ANSWER**

`ls -la` lists all files including hidden ones (names starting with `.`), with permissions, owner, size, and timestamps. `ls -lh` shows human-readable sizes.

INTERVIEW TIP

Explain each column of `ls -l` output if asked.

POSSIBLE FOLLOW-UP

What does the first character of `ls -l` output mean?

Q14. How do you create and remove directories?**BASIC****SHORT ANSWER**

`mkdir dir` creates a directory; `mkdir -p a/b/c` creates nested paths. `rmdir` removes empty directories; `rm -rf` removes directories with contents (use with caution).

INTERVIEW TIP

Say 'rm -rf is dangerous; I double-check the path and never run it as root casually'.

POSSIBLE FOLLOW-UP

What does `mkdir -p` do if the directory exists?

Q15. What is the difference between cp and mv?**BASIC****SHORT ANSWER**

cp copies files/directories (source remains); mv moves or renames them (source is removed). mv within the same filesystem only updates directory entries, so it is instant.

INTERVIEW TIP

Mention `cp -a` preserves permissions, ownership, and timestamps.

POSSIBLE FOLLOW-UP

How do you copy a directory recursively?

Q16. How do you view a file without opening an editor?**BASIC****SHORT ANSWER**

cat prints the whole file; less/more paginate; head -n 20 shows the first 20 lines; tail -n 20 the last 20; tail -f follows a growing log in real time.

INTERVIEW TIP

tail -f (or tail -F) on logs is a daily sysadmin task — say so.

POSSIBLE FOLLOW-UP

What is the difference between tail -f and tail -F?

Q17. What are file permissions in Linux?**BASIC****SHORT ANSWER**

Each file has read (r=4), write (w=2), execute (x=1) permissions for user, group, and others — e.g., 755 = rwxr-xr-x. chmod changes permissions, chown changes owner/group.

INTERVIEW TIP

Practice converting numeric to symbolic instantly (644, 755, 600).

POSSIBLE FOLLOW-UP

What does chmod 4755 do?

Q18. What does chmod 755 mean?**BASIC****SHORT ANSWER**

Owner: read+write+execute (7); group: read+execute (5); others: read+execute (5). Typical for scripts and directories.

INTERVIEW TIP

For directories, execute means 'can enter/traverse'.

POSSIBLE FOLLOW-UP

Why do directories need execute permission?

Q19. What is umask?**BASIC****SHORT ANSWER**

umask defines default permissions subtracted from the maximum (666 for files, 777 for directories). umask 022 gives new files 644 and directories 755.

INTERVIEW TIP

Compute an example live: $777 - 022 = 755$.

POSSIBLE FOLLOW-UP

Where do you set a permanent umask?

Q20. How do you change file ownership?**BASIC****SHORT ANSWER**

chown user:group file changes owner and group; chgrp changes group only; chown -R applies recursively.

INTERVIEW TIP

Only root (or via sudo) can change ownership to another user.

POSSIBLE FOLLOW-UP

Can a normal user chown their file to someone else?

Q21. What is the difference between su and sudo?**BASIC****SHORT ANSWER**

su switches to another user (needs that user's password, full session). sudo runs a single command with elevated privileges using your own password, controlled and logged via /etc/sudoers.

INTERVIEW TIP

Say 'sudo gives auditing and granular control — preferred in production'.

POSSIBLE FOLLOW-UP

How do you grant a user passwordless sudo for one command?

Q22. What is /etc/passwd?**BASIC****SHORT ANSWER**

A world-readable file listing user accounts: username, x (password placeholder), UID, GID, comment, home directory, and login shell — one line per user.

INTERVIEW TIP

Passwords are NOT here; they are hashed in /etc/shadow.

POSSIBLE FOLLOW-UP

Why is the password field just an 'x'?

Q23. What is /etc/shadow?**BASIC****SHORT ANSWER**

It stores hashed user passwords with aging info (last change, min/max days, expiry). Readable only by root, keeping hashes safe from normal users.

INTERVIEW TIP

Mention SHA-512 hashes (\$6\$) as the common default.

POSSIBLE FOLLOW-UP

How do you lock a user account?

Q24. How do you add a new user?**BASIC****SHORT ANSWER**

useradd -m -s /bin/bash username creates the user with a home directory and shell; passwd username sets the password. Debian systems also offer the interactive adduser.

INTERVIEW TIP

Mention -m explicitly — forgetting home dir creation is a common mistake.

POSSIBLE FOLLOW-UP

What files are copied into a new home directory and from where?

Q25. How do you add a user to a group?**BASIC****SHORT ANSWER**

usermod -aG groupname username appends the user to a supplementary group. Without -a, -G replaces all existing supplementary groups — a classic mistake.

INTERVIEW TIP

Highlight the -a flag pitfall; interviewers love it.

POSSIBLE FOLLOW-UP

How does the user get the new group without re-logging in?

Q26. What is the difference between primary and secondary groups?**BASIC****SHORT ANSWER**

The primary group (in /etc/passwd, GID field) owns files the user creates by default. Secondary/supplementary groups (in /etc/group) grant additional access rights.

INTERVIEW TIP

newgrp can switch the effective primary group temporarily.

POSSIBLE FOLLOW-UP

Where is the primary group defined?

Q27. How do you check which user you are and your groups?**BASIC****SHORT ANSWER**

whoami shows the current user; id shows UID, GID, and all groups; groups lists group memberships.

INTERVIEW TIP

id is more complete — prefer it in answers.

POSSIBLE FOLLOW-UP

What is the difference between real and effective UID?

Q28. What is a process?**BASIC****SHORT ANSWER**

A process is a running instance of a program with its own PID, memory space, and state. Processes are created by fork()/exec() and form a parent-child tree starting from PID 1.

INTERVIEW TIP

Mention PID 1 is systemd/init — the ancestor of all processes.

POSSIBLE FOLLOW-UP

What is the difference between a process and a thread?

Q29. How do you list running processes?**BASIC****SHORT ANSWER**

ps -ef or ps aux lists all processes with PID, owner, CPU/memory usage, and command. top/htop show a live, sortable view.

INTERVIEW TIP

Know ps aux column meanings (VSZ vs RSS).

POSSIBLE FOLLOW-UP

What is the difference between VSZ and RSS?

Q30. How do you kill a process?**BASIC****SHORT ANSWER**

kill PID sends SIGTERM (15, graceful). kill -9 PID sends SIGKILL (forced, cannot be trapped). pkill/killall target processes by name.

INTERVIEW TIP

Say 'I always try SIGTERM first; SIGKILL skips cleanup and can corrupt data'.

POSSIBLE FOLLOW-UP

Why can a process ignore SIGTERM but not SIGKILL?

Q31. What are foreground and background jobs?**BASIC****SHORT ANSWER**

A foreground job holds the terminal; appending & runs a command in the background. Ctrl+Z suspends a job; bg resumes it in background, fg brings it back; jobs lists them.

INTERVIEW TIP

Mention nohup/setsid or tmux for keeping jobs alive after logout.

POSSIBLE FOLLOW-UP

What happens to background jobs when you close the terminal?

Q32. What does the nohup command do?**BASIC****SHORT ANSWER**

nohup command & runs a command immune to hangup (SIGHUP), so it keeps running after the terminal closes. Output goes to nohup.out by default.

INTERVIEW TIP

In modern setups, prefer systemd services or tmux for long-running tasks.

POSSIBLE FOLLOW-UP

What is the difference between nohup and disown?

Q33. What is an environment variable?**BASIC****SHORT ANSWER**

A named value inherited by child processes — e.g., PATH, HOME, LANG. Set with export VAR=value; view with env or echo \$VAR. Persist in ~/.bashrc or /etc/profile.

INTERVIEW TIP

Explain the difference between shell variables and exported variables.

POSSIBLE FOLLOW-UP

What is the purpose of the PATH variable?

Q34. What is the PATH variable?**BASIC****SHORT ANSWER**

PATH is a colon-separated list of directories the shell searches for executables. If a command isn't in PATH, you must use its full path or add its directory to PATH.

INTERVIEW TIP

'command not found' usually means a PATH issue — say so.

POSSIBLE FOLLOW-UP

Why is '.' not in PATH by default?

Q35. What is the difference between > and >> ?

BASIC

SHORT ANSWER

> redirects stdout to a file, overwriting it; >> appends. 2> redirects stderr; &> or 2>&1 redirects both streams.

INTERVIEW TIP

Write `'command > out.log 2>&1'` from memory — a standard interview check.

POSSIBLE FOLLOW-UP

What does 2>&1 mean exactly?

Q36. What is a pipe (|)?

BASIC

SHORT ANSWER

A pipe connects the stdout of one command to the stdin of the next, letting you chain tools: `ps -ef | grep nginx | wc -l`.

INTERVIEW TIP

Give a real chained example, not just the definition.

POSSIBLE FOLLOW-UP

What is a named pipe (FIFO)?

Q37. How do you search text inside files?

BASIC

SHORT ANSWER

grep pattern file searches for a pattern; -i ignores case, -r searches recursively, -v inverts, -n shows line numbers, -c counts matches.

INTERVIEW TIP

`grep -r 'ERROR' /var/log` is a daily-use example worth quoting.

POSSIBLE FOLLOW-UP

What is the difference between grep, egrep, and fgrep?

Q38. How do you find files on the system?

BASIC

SHORT ANSWER

find /path -name '*.log' searches live by name, size (-size +100M), time (-mtime -1), type, or permissions, and can act with -exec. locate uses a prebuilt database (updatedb) — faster but possibly stale.

INTERVIEW TIP

find with -exec or piped to xargs is the production pattern.

POSSIBLE FOLLOW-UP

How do you find files larger than 100MB modified in the last day?

Q39. What is the difference between locate and find?

BASIC

SHORT ANSWER

locate queries a nightly-updated database (mlocate/plocate) — very fast but can be outdated. find scans the filesystem in real time — slower but always accurate and far more powerful.

INTERVIEW TIP

Mention updatedb refreshes the locate database.

POSSIBLE FOLLOW-UP

When would locate miss a file that exists?

Q40. How do you check disk usage?**BASIC****SHORT ANSWER**

`df -h` shows filesystem-level free/used space per mount. `du -sh /path` shows the size of a directory tree. `du -sh /*` helps find what is consuming space.

INTERVIEW TIP

df vs du mismatch (deleted-but-open files) is a favourite follow-up — be ready.

POSSIBLE FOLLOW-UP

Why can `df` show a full disk while `du` shows less usage?

Q41. How do you check memory usage?**BASIC****SHORT ANSWER**

`free -h` shows total, used, free, buffers/cache, and available memory plus swap. 'available' is the realistic figure for new applications.

INTERVIEW TIP

Explain that buff/cache is reclaimable — low 'free' is normal on Linux.

POSSIBLE FOLLOW-UP

Why does Linux show almost no free memory on a healthy server?

Q42. How do you check CPU and system load?**BASIC****SHORT ANSWER**

`top` or `htop` shows live CPU per process; `uptime` and `w` show load average (1, 5, 15 min). Load roughly equals runnable + uninterruptible tasks.

INTERVIEW TIP

Relate load to CPU count: load 4 on 4 cores = fully busy.

POSSIBLE FOLLOW-UP

What load average is 'high' on an 8-core server?

Q43. What is the tar command used for?**BASIC****SHORT ANSWER**

`tar` archives multiple files into one. `tar -czvf a.tar.gz dir/` creates a gzip archive; `tar -xzf a.tar.gz` extracts; `-tvf` lists contents; `-j` uses bzip2, `-J` uses xz.

INTERVIEW TIP

Recite create/extract/list syntax without hesitation.

POSSIBLE FOLLOW-UP

What is the difference between `.tar`, `.tar.gz`, and `.zip`?

Q44. What is the difference between gzip and zip?**BASIC****SHORT ANSWER**

`gzip` compresses a single file (usually paired with `tar` for archives) with better Unix integration; `zip` both archives and compresses, and is more portable to Windows.

INTERVIEW TIP

gzip removes the original file by default — mention `gzip -k` to keep it.

POSSIBLE FOLLOW-UP

How do you view a gzipped log without extracting it?

Q45. What are wget and curl used for?**BASIC****SHORT ANSWER**

Both fetch data over HTTP/HTTPS/FTP. wget is download-focused (recursive, resume with -c). curl is a general data-transfer tool supporting many protocols, headers, methods — ideal for API testing.

INTERVIEW TIP

curl -I for headers and curl -o for saving are handy examples.

POSSIBLE FOLLOW-UP

How do you test an API endpoint returning JSON with curl?

Q46. How do you check your IP address?**BASIC****SHORT ANSWER**

ip addr show (or ip a) lists interfaces and IPs — the modern command. ifconfig is deprecated. hostname -I prints only the IPs.

INTERVIEW TIP

Use ip, not ifconfig — interviewers note outdated habits.

POSSIBLE FOLLOW-UP

What replaced netstat and ifconfig in modern Linux?

Q47. How do you test connectivity to a host?**BASIC****SHORT ANSWER**

ping host checks ICMP reachability and latency. If ping is blocked, test the actual port: curl -v telnet://host:443, nc -zv host 443, or timeout 3 bash -c '</dev/tcp/host/443'.

INTERVIEW TIP

Say 'ping failing doesn't mean the service is down — ICMP may be blocked'.

POSSIBLE FOLLOW-UP

How do you check if a specific port is open on a remote server?

Q48. What is SSH?**BASIC****SHORT ANSWER**

SSH (Secure Shell) provides encrypted remote login and command execution over port 22. Authentication uses passwords or, preferably, public/private key pairs. Server config lives in /etc/ssh/sshd_config.

INTERVIEW TIP

Say 'we enforce key-based auth and disable root login'.

POSSIBLE FOLLOW-UP

How does SSH key-based authentication work?

Q49. How do you copy files between servers?**BASIC****SHORT ANSWER**

scp file user@host:/path copies over SSH. rsync -avz source user@host:/dest is preferred for large or repeated transfers — it is incremental, resumable, and preserves attributes.

INTERVIEW TIP

Position rsync as the production choice; scp for quick one-offs.

POSSIBLE FOLLOW-UP

What makes rsync faster than scp on repeated syncs?

Q50. What is cron?**BASIC****SHORT ANSWER**

cron runs scheduled jobs. `crontab -e` edits a user's jobs; `crontab -l` lists them. Format: minute hour day-of-month month day-of-week command. System-wide jobs live in `/etc/crontab` and `/etc/cron.d/`.

INTERVIEW TIP

Write `'0 2 * * * /backup.sh'` and explain it — runs daily at 02:00.

POSSIBLE FOLLOW-UP

How do you schedule a job every 5 minutes?

Q51. How do you check system logs?**BASIC****SHORT ANSWER**

On systemd systems: `journalctl` (e.g., `journalctl -u nginx -f`). Traditional files live in `/var/log` — `/var/log/messages` or `/var/log/syslog`, `/var/log/secure` or `auth.log`, plus app-specific logs.

INTERVIEW TIP

Give both `journalctl` and `/var/log` paths to cover RHEL and Debian.

POSSIBLE FOLLOW-UP

How do you view only errors from the last hour in `journalctl`?

Q52. What is the hostname and how do you change it?**BASIC****SHORT ANSWER**

The hostname identifies the machine on a network. `hostnamectl set-hostname web01` changes it persistently on systemd distros; also verify `/etc/hosts` entries.

INTERVIEW TIP

`hostnamectl` is the modern persistent method — `/etc/hostname` editing is the fallback.

POSSIBLE FOLLOW-UP

What is the difference between static, transient, and pretty hostnames?

Q53. What is /etc/hosts?**BASIC****SHORT ANSWER**

A local file mapping hostnames to IPs, consulted before DNS by default (per `/etc/nsswitch.conf`). Useful for overrides, testing, and small static setups.

INTERVIEW TIP

Mention `nsswitch.conf` decides `hosts-vs-DNS` order.

POSSIBLE FOLLOW-UP

How do you make DNS take priority over `/etc/hosts`?

Q54. How do you check which shell and OS version you're running?**BASIC****SHORT ANSWER**

`echo $SHELL` or `echo $0` shows the shell. `cat /etc/os-release` shows distro and version; `uname -r` shows the kernel version; `uname -a` shows full kernel info.

INTERVIEW TIP

`cat /etc/os-release` works across all modern distros — safest answer.

POSSIBLE FOLLOW-UP

How do you check the kernel version only?

Q55. What does the history command do?**BASIC****SHORT ANSWER**

history lists previously executed commands (stored in `~/.bash_history`). It re-runs command `n`; `!!` repeats the last command; `Ctrl+R` searches history interactively.

INTERVIEW TIP

Mention `HISTSIZE/HISTTIMEFORMAT` for audit-friendly history.

POSSIBLE FOLLOW-UP

How do you add timestamps to bash history?

Q56. What is the alias command?**BASIC****SHORT ANSWER**

alias creates command shortcuts, e.g., `alias ll='ls -la'`. Defined per-session or persisted in `~/.bashrc`. `unalias` removes one.

INTERVIEW TIP

Warn: aliases can mask real commands in scripts; scripts don't load interactive aliases.

POSSIBLE FOLLOW-UP

Why don't aliases work inside shell scripts by default?

Q57. What is the difference between reboot, shutdown, and halt?**BASIC****SHORT ANSWER**

`shutdown -r now` reboots gracefully; `shutdown -h now` powers off; `halt` stops the CPU without necessarily powering off. `systemctl reboot/poweroff` are the `systemd` equivalents.

INTERVIEW TIP

`shutdown -r +5 'message'` warns users — mention graceful production practice.

POSSIBLE FOLLOW-UP

How do you cancel a scheduled shutdown?

Q58. What are stdin, stdout, and stderr?**BASIC****SHORT ANSWER**

They are the three standard streams: `stdin` (fd 0) input, `stdout` (fd 1) normal output, `stderr` (fd 2) error output. They can be redirected independently, which is why `2>&1` exists.

INTERVIEW TIP

Tie it back to redirection syntax immediately.

POSSIBLE FOLLOW-UP

How do you discard errors but keep output?

Q59. What is the touch command for?**BASIC****SHORT ANSWER**

`touch` creates an empty file if it doesn't exist, or updates the access/modification timestamps of an existing file (`-a`, `-m`, `-t` for specific times).

INTERVIEW TIP

Timestamp updating is the 'senior' half of the answer — don't skip it.

POSSIBLE FOLLOW-UP

How do you set a file's timestamp to a specific date?

Q60. What is the man command?**BASIC****SHORT ANSWER**

man shows the manual page for a command (man ls). Sections matter: man 5 passwd shows the file format vs man 1 passwd the command. apropos/man -k searches by keyword.

INTERVIEW TIP

Knowing man sections (1 commands, 5 file formats, 8 admin) impresses.

POSSIBLE FOLLOW-UP

What is the difference between man 1 crontab and man 5 crontab?

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SECTION 2

Intermediate Linux Questions

Daily system administration: systemd, storage, networking, security

Q61. What is systemd?

INTERMEDIATE

SHORT ANSWER

systemd is the init system (PID 1) on modern distros. It boots the system, manages services as units with dependencies, supports parallel startup, sockets, timers, and integrates logging via journald.

INTERVIEW TIP

Contrast with SysV init: parallel startup, dependency handling, unified tooling.

POSSIBLE FOLLOW-UP

What are systemd targets and how do they map to runlevels?

Q62. What are common systemctl commands?

INTERMEDIATE

SHORT ANSWER

systemctl start|stop|restart|reload|status unit; enable/disable for boot behavior; is-active/is-enabled for checks; daemon-reload after editing unit files; list-units --failed for failures.

INTERVIEW TIP

Clarify enable ≠ start: enable only affects next boot.

POSSIBLE FOLLOW-UP

What is the difference between restart and reload?

Q63. What is a systemd unit file?

INTERMEDIATE

SHORT ANSWER

A unit file describes something systemd manages (service, timer, mount, socket). Service files have [Unit] (description, dependencies), [Service] (ExecStart, Restart, User), and [Install] (WantedBy) sections. Custom units go in /etc/systemd/system/.

INTERVIEW TIP

Say you override vendor units with systemctl edit (drop-ins), not by editing /usr/lib files.

POSSIBLE FOLLOW-UP

How do you make a service auto-restart on failure?

Q64. How does journalctl work?

INTERMEDIATE

SHORT ANSWER

journalctl queries systemd's binary journal. Key flags: -u unit, -f follow, -b current boot, --since '1 hour ago', -p err for priority filtering. Persistent storage needs /var/log/journal.

INTERVIEW TIP

journalctl -u app -p err --since today is a great one-liner to quote.

POSSIBLE FOLLOW-UP

How do you make the journal persistent across reboots?

Q65. Explain the Linux boot process.**INTERMEDIATE****SHORT ANSWER**

BIOS/UEFI runs POST and loads the bootloader (GRUB2) → GRUB loads the kernel and initramfs → kernel initializes hardware, mounts the initramfs then the real root filesystem → starts systemd (PID 1) → systemd reaches default.target, starting services.

INTERVIEW TIP

Deliver it as a confident 5-step story; this is a guaranteed question.

POSSIBLE FOLLOW-UP

What is the role of initramfs?

Q66. What is GRUB and where is its configuration?**INTERMEDIATE****SHORT ANSWER**

GRUB2 is the bootloader that presents the boot menu and loads the kernel. Config is generated into /boot/grub2/grub.cfg (BIOS) or /boot/efi/... (UEFI) from /etc/default/grub via grub2-mkconfig / update-grub. Never edit grub.cfg directly.

INTERVIEW TIP

Mention editing /etc/default/grub then regenerating — direct edits are a red flag.

POSSIBLE FOLLOW-UP

How do you recover a server with a corrupted GRUB?

Q67. What is initramfs?**INTERMEDIATE****SHORT ANSWER**

A compressed, temporary root filesystem loaded into RAM by the bootloader. It contains drivers/modules (storage, LVM, RAID) needed to mount the real root filesystem, then hands over control. Rebuilt with dracut or update-initramfs.

INTERVIEW TIP

'Kernel panics after adding storage drivers' often means initramfs rebuild needed.

POSSIBLE FOLLOW-UP

When must you regenerate the initramfs?

Q68. What are runlevels/targets?**INTERMEDIATE****SHORT ANSWER**

systemd targets replaced runlevels: multi-user.target (runlevel 3, non-GUI server) and graphical.target (runlevel 5) are most common. systemctl get-default shows the default; set-default changes it; isolate switches live.

INTERVIEW TIP

rescue.target and emergency.target for recovery — mention them.

POSSIBLE FOLLOW-UP

How do you boot into single-user/rescue mode?

Q69. What are zombie and orphan processes?

INTERMEDIATE

SHORT ANSWER

A zombie is a terminated process whose exit status hasn't been reaped by its parent — it holds only a process-table entry. An orphan's parent died; it is re-parented to PID 1 which reaps it. Many zombies indicate a buggy parent.

INTERVIEW TIP

You cannot kill a zombie — kill/fix or restart the parent.

POSSIBLE FOLLOW-UP

How do you find and clear zombie processes?

Q70. What is nice and renice?

INTERMEDIATE

SHORT ANSWER

Niceness (-20 to 19) adjusts scheduling priority: lower = higher priority. `nice -n 10 cmd` starts with lower priority; `renice -n -5 -p PID` changes a running process (negative values need root).

INTERVIEW TIP

Use case: renice a heavy batch job so it doesn't starve the app.

POSSIBLE FOLLOW-UP

What is the difference between nice value and real-time priority?

Q71. What are signals in Linux?

INTERMEDIATE

SHORT ANSWER

Signals are async notifications to processes: SIGTERM(15) graceful stop, SIGKILL(9) force kill, SIGHUP(1) hangup/reload config, SIGINT(2) Ctrl+C, SIGSTOP/SIGCONT pause/resume. `kill -l` lists all.

INTERVIEW TIP

SIGHUP for config reload (nginx, sshd) is a strong practical detail.

POSSIBLE FOLLOW-UP

Why can't SIGKILL and SIGSTOP be caught or ignored?

Q72. How do you run a command at a specific time once?

INTERMEDIATE

SHORT ANSWER

The `at` command: `echo '/script.sh' | at 22:00` schedules a one-time job; `atq` lists queue, `atrm` removes. For recurring jobs, use `cron` or `systemd` timers.

INTERVIEW TIP

Contrast at (one-time) vs cron (recurring) crisply.

POSSIBLE FOLLOW-UP

What are `systemd` timers and why prefer them over `cron`?

Q73. Explain the crontab format with an example.

INTERMEDIATE

SHORT ANSWER

Five fields then the command: minute hour day-of-month month day-of-week. `*/* 5 * * *` = every 5 min; `'0 2 * * 0'` = 02:00 every Sunday. Special strings: `@reboot`, `@daily`. User crontabs via `crontab -e`; system ones in `/etc/cron.d/` include a user field.

INTERVIEW TIP

Warn: cron has a minimal PATH — always use absolute paths in jobs.

POSSIBLE FOLLOW-UP

Why does a script work manually but fail in `cron`?

Q74. What is the difference between APT and YUM/DNF?**INTERMEDIATE****SHORT ANSWER**

APT (apt, dpkg, .deb) serves Debian/Ubuntu; YUM/DNF (rpm, .rpm) serve RHEL/Fedora/CentOS. Both resolve dependencies from repositories. DNF replaced YUM with better dependency resolution and performance.

INTERVIEW TIP

Map: `dpkg↔rpm` (low-level), `apt↔dnf` (high-level).

POSSIBLE FOLLOW-UP

How do you list which package owns a specific file?

Q75. How do you install, query, and remove packages on RHEL?**INTERMEDIATE****SHORT ANSWER**

`dnf install pkg`, `dnf remove pkg`, `dnf update`; `rpm -qa` lists installed, `rpm -qf /path` finds owning package, `rpm -ql pkg` lists its files; `dnf provides */filename` finds candidates; `dnf history` shows/rolls back transactions.

INTERVIEW TIP

`dnf history undo` is a production-saver — mention it.

POSSIBLE FOLLOW-UP

How do you downgrade a package safely?

Q76. What is a repository and how do you add one?**INTERMEDIATE****SHORT ANSWER**

A repo is a server hosting packages plus metadata. RHEL: .repo files in `/etc/yum.repos.d/` with `baseurl` and `gpgcheck`. Debian: entries in `/etc/apt/sources.list(.d)` then `apt update`. GPG keys verify package integrity.

INTERVIEW TIP

Mention `gpgcheck=1` and key verification — security awareness scores points.

POSSIBLE FOLLOW-UP

What happens if GPG check fails during installation?

Q77. How does SSH key-based authentication work?**INTERMEDIATE****SHORT ANSWER**

You generate a key pair (`ssh-keygen`). The public key goes into the server's `~/.ssh/authorized_keys` (`ssh-copy-id`). At login the server issues a challenge only your private key can sign, so no password crosses the network.

INTERVIEW TIP

Mention permissions: `~/.ssh 700`, `authorized_keys 600` — wrong perms silently break it.

POSSIBLE FOLLOW-UP

SSH key auth suddenly fails — what do you check first?

Q78. What are common sshd_config hardening options?**INTERMEDIATE****SHORT ANSWER**

`PermitRootLogin no`, `PasswordAuthentication no` (keys only), `AllowUsers/AllowGroups`, Port change if policy requires, `MaxAuthTries`, `ClientAliveInterval` for idle timeout. Reload with `systemctl reload sshd` after testing `sshd -t`.

INTERVIEW TIP

Always say `'sshd -t before reload, and keep one session open while testing'`.

POSSIBLE FOLLOW-UP

How do you restrict SSH to a specific group of users?

Q79. What is SSH tunneling / port forwarding?**INTERMEDIATE****SHORT ANSWER**

Local forwarding (`ssh -L 8080:db:5432 user@bastion`) exposes a remote service locally; remote forwarding (`-R`) does the reverse; dynamic (`-D`) creates a SOCKS proxy. Common for reaching DBs behind bastions.

INTERVIEW TIP

The bastion/jump-host use case (also `ssh -J`) is the expected real-world example.

POSSIBLE FOLLOW-UP

What is the difference between `-L` and `-R` forwarding?

Q80. What is rsync and its key options?**INTERMEDIATE****SHORT ANSWER**

`rsync` synchronizes files incrementally using delta transfer. `rsync -avz src/ user@host:/dest/` — `-a` preserves attributes, `-v` verbose, `-z` compress; `--delete` mirrors deletions; `--dry-run` previews; `--exclude` filters.

INTERVIEW TIP

Explain the trailing-slash semantics on the source path — a classic gotcha.

POSSIBLE FOLLOW-UP

What does a trailing slash on the `rsync` source change?

Q81. How do you check open ports and listening services?**INTERMEDIATE****SHORT ANSWER**

`ss -tulnp` lists TCP/UDP listening sockets with owning processes (`netstat -tulnp` is the legacy equivalent). `lsof -i :80` shows what holds port 80.

INTERVIEW TIP

Use `ss` first; note `netstat` is deprecated.

POSSIBLE FOLLOW-UP

How do you find which process is using port 8080?

Q82. What is the difference between TCP and UDP?**INTERMEDIATE****SHORT ANSWER**

TCP is connection-oriented and reliable: handshake, ordering, retransmission (HTTP, SSH, DB). UDP is connectionless and fast with no delivery guarantee (DNS queries, NTP, streaming).

INTERVIEW TIP

Tie each to a service example — that's what interviewers want.

POSSIBLE FOLLOW-UP

Why does DNS use UDP but switch to TCP sometimes?

Q83. How does DNS resolution work on a Linux host?**INTERMEDIATE****SHORT ANSWER**

The resolver checks `/etc/nsswitch.conf` order — typically `/etc/hosts` first, then DNS servers from `/etc/resolv.conf` (often managed by `systemd-resolved` or `NetworkManager`). Tools: `dig`, `nslookup`, `resolvectl`.

INTERVIEW TIP

`dig +short` and `dig @8.8.8.8` to isolate resolver vs record issues — quote them.

POSSIBLE FOLLOW-UP

How do you test DNS against a specific nameserver?

Q84. What is the purpose of the ip command?**INTERMEDIATE****SHORT ANSWER**

ip replaces ifconfig/route/arp: ip addr (addresses), ip link (interfaces up/down), ip route (routing table), ip neigh (ARP). Example: ip route add default via 192.168.1.1.

INTERVIEW TIP

Changes via ip are non-persistent — mention persisting via nmcli/netplan/ifcfg.

POSSIBLE FOLLOW-UP

How do you make network changes persistent across reboots?

Q85. What is a default gateway and how do you view/change it?**INTERMEDIATE****SHORT ANSWER**

The default gateway is the router used for traffic outside local subnets. ip route show displays it ('default via ...'); ip route replace default via X.X.X.X changes it at runtime.

INTERVIEW TIP

A missing default route explains 'can ping LAN, not internet' — use that scenario.

POSSIBLE FOLLOW-UP

Server can reach local subnet but not the internet — what do you check?

Q86. What is firewalld and its basic usage?**INTERMEDIATE****SHORT ANSWER**

firewalld is the dynamic firewall manager on RHEL-family systems using zones. firewall-cmd --list-all shows rules; --add-service=https or --add-port=8080/tcp opens access; add --permanent then --reload to persist.

INTERVIEW TIP

The permanent+reload two-step is the detail that shows real usage.

POSSIBLE FOLLOW-UP

What is the difference between runtime and permanent firewalld rules?

Q87. How do iptables rules work?**INTERMEDIATE****SHORT ANSWER**

iptables filters packets through tables (filter, nat, mangle) and chains (INPUT, OUTPUT, FORWARD). Rules match in order; the first match wins, otherwise chain policy applies. Example: iptables -A INPUT -p tcp --dport 22 -j ACCEPT.

INTERVIEW TIP

Mention rule order matters and nftables is the modern backend.

POSSIBLE FOLLOW-UP

What is the difference between DROP and REJECT?

Q88. What is SELinux?**INTERMEDIATE****SHORT ANSWER**

SELinux enforces mandatory access control with policies and contexts on processes/files, beyond ownership/permissions. Modes: Enforcing, Permissive (log only), Disabled. Tools: getenforce, sestatus, ls -Z, restorecon, semanage, audit2why.

INTERVIEW TIP

Say 'we keep it Enforcing and fix contexts, we don't disable it' — big credibility boost.

POSSIBLE FOLLOW-UP

A service can't read files despite correct permissions — how does SELinux fit in?

Q89. How do you troubleshoot an SELinux denial?**INTERMEDIATE****SHORT ANSWER**

Check /var/log/audit/audit.log or ausearch -m AVC -ts recent; interpret with audit2why. Fix contexts with restorecon -Rv /path or set them with semanage fcontext; toggle feature booleans with setsebool -P.

INTERVIEW TIP

restorecon after moving files into web roots is the classic real fix.

POSSIBLE FOLLOW-UP

Why does mv into /var/www break SELinux contexts but cp doesn't?

Q90. What is the difference between a disk, partition, and filesystem?**INTERMEDIATE****SHORT ANSWER**

A disk (/dev/sda, /dev/nvme0n1) is raw storage; partitions (/dev/sda1) divide it (MBR or GPT tables); a filesystem (ext4, xfs) formats a partition/volume to store files, then gets mounted to a directory.

INTERVIEW TIP

State the workflow: partition → mkfs → mount → fstab.

POSSIBLE FOLLOW-UP

What is the difference between MBR and GPT?

Q91. How do you add a new disk to a Linux server?**INTERMEDIATE****SHORT ANSWER**

Scan/verify with lsblk; partition with parted/fdisk (GPT preferred); mkfs.xfs /dev/sdb1; mkdir /data && mount /dev/sdb1 /data; add an fstab entry using UUID (blkid) and validate with mount -a.

INTERVIEW TIP

Always say 'UUID in fstab, then mount -a to test before reboot'.

POSSIBLE FOLLOW-UP

Why should you use UUIDs instead of device names in fstab?

Q92. What is /etc/fstab?**INTERMEDIATE****SHORT ANSWER**

fstab defines filesystems mounted at boot: device (UUID), mount point, fs type, options (defaults, noatime), dump, and fsck order. Errors here can drop the server into emergency mode at boot.

INTERVIEW TIP

Mention nofail option for non-critical disks to avoid boot hangs.

POSSIBLE FOLLOW-UP

A bad fstab entry sent the server to emergency mode — how do you recover?

Q93. What is swap and when is it used?**INTERMEDIATE****SHORT ANSWER**

Swap is disk space used when RAM is scarce; the kernel pushes inactive pages there. `vm.swappiness` (0–100) tunes eagerness. Create via a swap partition or file (`mkswap`, `swapon`), persist in `fstab`.

INTERVIEW TIP

Heavy swapping = performance killer; monitor `si/so` in `vmstat`.

POSSIBLE FOLLOW-UP

How do you add swap space on a running server without a new partition?

Q94. What is LVM and its main components?**INTERMEDIATE****SHORT ANSWER**

LVM adds a flexible layer over disks: Physical Volumes (PV) are grouped into Volume Groups (VG), from which Logical Volumes (LV) are carved. Benefits: online resizing, snapshots, spanning multiple disks.

INTERVIEW TIP

Draw the `PV→VG→LV` hierarchy verbally in one sentence.

POSSIBLE FOLLOW-UP

How do you extend a logical volume and its filesystem online?

Q95. How do you extend an LVM filesystem online?**INTERMEDIATE****SHORT ANSWER**

Add/extend a PV (`pvcreate`, `vgextend`), then `lvextend -r -L +10G /dev/vg/lv` — the `-r` resizes the filesystem too (`xfs_growfs/resize2fs` under the hood). No downtime for `ext4/xfs` growth.

INTERVIEW TIP

Remember: XFS can grow but never shrink.

POSSIBLE FOLLOW-UP

Can you shrink an XFS filesystem?

Q96. What is the mount command and how do you unmount a busy filesystem?**INTERMEDIATE****SHORT ANSWER**

`mount` attaches filesystems (`mount /dev/sdb1 /data`); `umount` detaches. If busy, find users with `lsof +f -- /data` or `fuser -vm /data`, stop them, or `umount -l` (lazy) as last resort.

INTERVIEW TIP

`fuser -km /mnt` kills users of the mount — powerful but say 'with caution'.

POSSIBLE FOLLOW-UP

What does 'target is busy' mean and how do you resolve it?

Q97. What is NFS and how do you mount a share?**INTERMEDIATE****SHORT ANSWER**

NFS shares directories over the network. Server exports in `/etc/exports` (`exportfs -ra`); client mounts with `mount -t nfs server:/share /mnt` and persists in `fstab`. `showmount -e server` lists exports.

INTERVIEW TIP

Mention hard vs soft mount options and their risk trade-offs.

POSSIBLE FOLLOW-UP

What is the difference between hard and soft NFS mounts?

Q98. What are SUID, SGID, and the sticky bit?**INTERMEDIATE****SHORT ANSWER**

SUID (4xxx): file runs with owner's privileges (e.g., passwd). SGID (2xxx): runs with group privileges; on directories, new files inherit the group. Sticky bit (1xxx) on directories: only owners can delete their files (e.g., /tmp = 1777).

INTERVIEW TIP

Give /usr/bin/passwd and /tmp as canonical examples.

POSSIBLE FOLLOW-UP

How do you find all SUID files on a system and why would you?

Q99. What are ACLs in Linux?**INTERMEDIATE****SHORT ANSWER**

Access Control Lists grant permissions beyond owner/group/other: setfacl -m u:john:rw file; view with getfacl. A '+' at the end of ls -l permissions indicates ACLs; default ACLs on directories set inheritance.

INTERVIEW TIP

Use case: granting one extra team read access without changing group ownership.

POSSIBLE FOLLOW-UP

What does the + sign in ls -l output indicate?

Q100. How do you check service failure reasons?**INTERMEDIATE****SHORT ANSWER**

systemctl status unit shows state and recent log lines; journalctl -u unit -xe gives detailed logs; check exit codes, config validity (e.g., nginx -t), port conflicts (ss -tlnp), permissions, and SELinux denials.

INTERVIEW TIP

Present it as a checklist — structured thinking is what's being tested.

POSSIBLE FOLLOW-UP

A service exits immediately after start — what are the top causes?

Q101. What is logrotate?**INTERMEDIATE****SHORT ANSWER**

logrotate rotates, compresses, and prunes logs on schedule via /etc/logrotate.conf and /etc/logrotate.d/. Options: rotate count, size/daily/weekly, compress, postrotate scripts. Test with logrotate -d (debug) or -f (force).

INTERVIEW TIP

Mention copytruncate vs reload for apps that keep files open.

POSSIBLE FOLLOW-UP

The app keeps writing to the rotated (deleted) file — why, and what fixes it?

Q102. How do you monitor disk I/O?**INTERMEDIATE****SHORT ANSWER**

iostat -x 2 shows per-device utilization, await, and throughput; iotop shows per-process I/O; vmstat shows blocked processes (b column) and wa CPU. High %util and await indicate saturation.

INTERVIEW TIP

Interpret metrics, don't just name tools: 'await up + %util ~100% = saturated disk'.

POSSIBLE FOLLOW-UP

What does high iowait actually indicate?

Q103. What do vmstat columns tell you?**INTERMEDIATE****SHORT ANSWER**

vmstat 2 shows r (runnable queue), b (blocked on I/O), si/so (swap in/out), bi/bo (block I/O), and CPU us/sy/id/wa. r > cores = CPU pressure; nonzero si/so = memory pressure; high wa = I/O bottleneck.

INTERVIEW TIP

Being able to read one vmstat line aloud is a strong differentiator.

POSSIBLE FOLLOW-UP

What does a consistently high 'b' column indicate?

Q104. What is sar and why use it?**INTERMEDIATE****SHORT ANSWER**

sar (sysstat) records historical performance data — CPU, memory, I/O, network — collected via cron. sar -u shows CPU history, sar -r memory, sar -n DEV network. Essential for analyzing issues that happened in the past.

INTERVIEW TIP

'The server was slow at 3 AM' → sar is the answer. Say exactly that.

POSSIBLE FOLLOW-UP

How do you check yesterday's CPU usage on a server?

Q105. What is the difference between grep, sed, and awk?**INTERMEDIATE****SHORT ANSWER**

grep searches/filters lines by pattern; sed is a stream editor for substitution and line edits (sed 's/old/new/g'); awk is a field-oriented processing language for extracting columns, computing, and reporting (awk '{print \$1}').

INTERVIEW TIP

One-line uses: grep filters, sed replaces, awk extracts columns.

POSSIBLE FOLLOW-UP

How do you print the 2nd column of a CSV with awk?

Q106. Give a practical sed example.**INTERMEDIATE****SHORT ANSWER**

sed -i 's/8080/9090/g' app.conf replaces all occurrences in place; sed -n '10,20p' file prints lines 10–20; sed '/^#/d' removes comment lines. Use -i.bak to keep a backup.

INTERVIEW TIP

Mention -i.bak — editing configs without backup is a red flag.

POSSIBLE FOLLOW-UP

How do you delete blank lines from a file with sed?

Q107. Give a practical awk example.**INTERMEDIATE****SHORT ANSWER**

awk -F: '{print \$1, \$7}' /etc/passwd prints usernames and shells; awk '\$9==500' access.log filters column matches; awk '{sum+=\$5} END{print sum}' totals a column.

INTERVIEW TIP

The /etc/passwd example is universally understood — lead with it.

POSSIBLE FOLLOW-UP

How do you sum values in a column using awk?

Q108. What do sort, uniq, cut, and xargs do?**INTERMEDIATE****SHORT ANSWER**

cut extracts fields (cut -d: -f1); sort orders lines (-n numeric, -r reverse, -k key); uniq deduplicates adjacent lines (-c counts) so it follows sort; xargs builds commands from stdin (find ... | xargs rm).

INTERVIEW TIP

Chain them: `sort | uniq -c | sort -nr` for a top-N frequency report.

POSSIBLE FOLLOW-UP

Why must you sort before using uniq?

Q109. What does the tee command do?**INTERMEDIATE****SHORT ANSWER**

tee writes stdin to both stdout and a file: `command | tee out.log` shows output live while saving it; -a appends. Common with sudo: `echo 'line' | sudo tee -a /etc/file`.

INTERVIEW TIP

The 'sudo tee' pattern for writing root-owned files is the interview-worthy detail.

POSSIBLE FOLLOW-UP

Why does 'sudo echo x > /etc/file' fail but 'echo x | sudo tee /etc/file' work?

Q110. What is the /dev directory and special files like /dev/null?**INTERMEDIATE****SHORT ANSWER**

/dev holds device files: block devices (sda, nvme0n1), character devices (tty). /dev/null discards writes; /dev/zero yields zeros; /dev/urandom yields random data. Redirecting to /dev/null silences output.

INTERVIEW TIP

command > /dev/null 2>&1 — recite it and explain both redirects.

POSSIBLE FOLLOW-UP

What is the difference between /dev/random and /dev/urandom?

SECTION 3

Advanced Linux Questions

Performance, internals, and recovery for senior roles

Q111. Explain load average correctly.**ADVANCED****SHORT ANSWER**

Load average is the average number of processes runnable or in uninterruptible sleep (D state, usually I/O) over 1/5/15 minutes. It must be read relative to CPU count: load 8 on 8 cores = saturation; on 32 cores it's light.

INTERVIEW TIP

Mention D-state inclusion — high load with idle CPU means I/O, not CPU, pressure.

POSSIBLE FOLLOW-UP

Load is 20 but CPU is 90% idle — what is happening?

Q112. What is iowait and what does high iowait mean?**ADVANCED****SHORT ANSWER**

iowait (%wa) is CPU time spent idle while at least one task waits on disk I/O. Persistently high iowait with high device await/%util in iostat indicates a storage bottleneck — slow disk, saturated volume, or heavy sync writes.

INTERVIEW TIP

Correlate wa (vmstat/top) with iostat -x await/%util — the pairing shows expertise.

POSSIBLE FOLLOW-UP

How do you find which process is causing the I/O load?

Q113. What is the OOM killer and how does it choose victims?**ADVANCED****SHORT ANSWER**

When memory plus swap is exhausted, the kernel's OOM killer terminates a process to survive. It picks by oom_score (memory footprint, adjusted by oom_score_adj). Kills are logged in dmesg/journal ('Out of memory: Killed process').

INTERVIEW TIP

Mention protecting critical daemons via OOMScoreAdjust in the systemd unit.

POSSIBLE FOLLOW-UP

How do you protect a critical process from the OOM killer?

Q114. How do you analyze memory usage deeply?**ADVANCED****SHORT ANSWER**

free -h for overview; /proc/meminfo for detail; smem or ps aux --sort=-rss for per-process RSS; /proc/PID/smmaps for a process's mappings; slabtop for kernel slab usage. Distinguish RSS (real) vs VSZ (virtual) and account for shared pages.

INTERVIEW TIP

Explain why summing RSS overcounts due to shared libraries.

POSSIBLE FOLLOW-UP

What is the difference between RSS, VSZ, and PSS?

Q115. What are huge pages?**ADVANCED****SHORT ANSWER**

Huge pages (2MB/1GB vs 4KB) reduce TLB misses and page-table overhead for large-memory apps (databases, JVMs). Configured via `vm.nr_hugepages`; Transparent Huge Pages (THP) does it automatically but is often disabled for databases due to latency spikes.

INTERVIEW TIP

'We disable THP for Oracle/MongoDB' is a well-known production practice to cite.

POSSIBLE FOLLOW-UP

Why do database vendors recommend disabling THP?

Q116. What is strace and when do you use it?**ADVANCED****SHORT ANSWER**

`strace` traces a process's system calls and signals: `strace -p PID` attaches live; `-f` follows children; `-e trace=open,connect` filters; `-c` summarizes counts/latency. Ideal for 'why does this hang/fail' — seeing the exact failing syscall and errno.

INTERVIEW TIP

A concrete story ('found ENOENT on a missing config via strace') beats theory.

POSSIBLE FOLLOW-UP

`strace` shows EACCES on a file open — what are the possible causes?

Q117. What is lsof and give practical uses.**ADVANCED****SHORT ANSWER**

`lsof` lists open files (everything is a file): `lsof -i :443` (port owner), `lsof /var/log/app.log` (who holds a file), `lsof +L1` (deleted-but-open files), `lsof -u user`. Crucial for busy mounts and disk-space mysteries.

INTERVIEW TIP

`lsof +L1` to explain `df/du` mismatch is the killer use case.

POSSIBLE FOLLOW-UP

`df` shows disk full but `du` doesn't add up — how does `lsof` help?

Q118. How does tcpdump help in troubleshooting?**ADVANCED****SHORT ANSWER**

`tcpdump` captures packets: `tcpdump -i eth0 port 443 -nn; -w file.pcap` saves for Wireshark; filters by host/port/proto. Use it to confirm whether traffic arrives, where handshakes fail (SYN with no SYN-ACK), and to inspect latency/retransmits.

INTERVIEW TIP

'SYN sent, no SYN-ACK = blocked upstream or dead host' — reason from packets.

POSSIBLE FOLLOW-UP

You see SYN packets but no SYN-ACK — what does that tell you?

Q119. Explain the TCP three-way handshake and TIME_WAIT.**ADVANCED****SHORT ANSWER**

Connection: SYN → SYN-ACK → ACK. Teardown leaves the closing side in TIME_WAIT (2×MSL) to absorb stray packets. Thousands of TIME_WAIT sockets are normal on busy servers; tune with net.ipv4.tcp_tw_reuse rather than panicking.

INTERVIEW TIP

Say 'TIME_WAIT is by design, not a fault' — many candidates get this wrong.

POSSIBLE FOLLOW-UP

A server has 30k TIME_WAIT sockets — is that a problem?

Q120. What is a CLOSE_WAIT socket pile-up?**ADVANCED****SHORT ANSWER**

CLOSE_WAIT means the peer closed but the local application hasn't called close() — a local application bug (leaked sockets/file descriptors). Growing CLOSE_WAIT counts eventually exhaust FDs; the fix is in the app, plus restart as mitigation.

INTERVIEW TIP

CLOSE_WAIT = our app's fault; TIME_WAIT = normal. That contrast wins the question.

POSSIBLE FOLLOW-UP

How do you identify which process holds CLOSE_WAIT sockets?

Q121. What are file descriptor limits and how do you raise them?**ADVANCED****SHORT ANSWER**

Each process has an FD limit (ulimit -n; default often 1024). Exhaustion causes 'Too many open files'. Raise per-user in /etc/security/limits.conf (nofile), system-wide via fs.file-max, and for services via LimitNOFILE= in the systemd unit.

INTERVIEW TIP

For systemd services, limits.conf does NOT apply — LimitNOFILE is the fix. Big differentiator.

POSSIBLE FOLLOW-UP

Why didn't editing limits.conf fix the limit for a systemd service?

Q122. What are cgroups?**ADVANCED****SHORT ANSWER**

Control groups limit and account resources (CPU, memory, I/O, pids) for groups of processes. systemd places every service in a cgroup, enabling MemoryMax=, CPUQuota= in units. cgroups v2 unifies the hierarchy; they underpin containers.

INTERVIEW TIP

Link cgroups+namespaces = containers — interviewers expect that connection.

POSSIBLE FOLLOW-UP

How do you cap a service at 2 CPUs and 4GB RAM using systemd?

Q123. What are Linux namespaces?**ADVANCED****SHORT ANSWER**

Namespaces isolate what processes see: PID, network, mount, UTS (hostname), IPC, user, cgroup, time. Each container gets its own set, so it has its own PIDs, interfaces, and mounts. Inspect with `lsns`; enter with `nsenter`.

INTERVIEW TIP

Naming `nsenter` for debugging containers from the host shows hands-on depth.

POSSIBLE FOLLOW-UP

How do namespaces and cgroups differ in purpose?

Q124. Explain the difference between processes and threads at the kernel level.**ADVANCED****SHORT ANSWER**

Linux creates both via `clone()`; threads share address space, FDs, and signal handlers (`CLONE_VM`, `CLONE_FILES`), while processes get copies (copy-on-write via `fork`). The scheduler treats both as tasks; threads of one process share a TGID.

INTERVIEW TIP

Mention copy-on-write fork — it explains why fork of a huge process is cheap.

POSSIBLE FOLLOW-UP

Why is `fork()` fast even for a process using 10GB RAM?

Q125. What is a kernel panic and how do you approach it?**ADVANCED****SHORT ANSWER**

A kernel panic is an unrecoverable kernel error halting the system. Causes: bad drivers/modules, corrupt `initramfs`, failing hardware/RAM, bad kernel update. Approach: capture the console/vmcore (`kdump`), boot a previous kernel from GRUB, rebuild `initramfs`, check hardware.

INTERVIEW TIP

'Boot the previous kernel from GRUB' is the expected first recovery step.

POSSIBLE FOLLOW-UP

What is `kdump` and why enable it?

Q126. What is kdump?**ADVANCED****SHORT ANSWER**

`kdump` reserves memory (`crashkernel=`) to boot a capture kernel on panic and dump memory (vmcore) to `/var/crash`. The vmcore is analyzed with the `crash` utility or sent to the vendor to identify the faulty module/driver.

INTERVIEW TIP

Enterprise support (Red Hat) asks for vmcore — mention that workflow.

POSSIBLE FOLLOW-UP

How do you verify `kdump` is properly configured?

Q127. What are sysctl parameters and common ones you've tuned?**ADVANCED****SHORT ANSWER**

sysctl reads/writes kernel parameters at runtime (/proc/sys). Persist in /etc/sysctl.d/*.conf; apply with sysctl -p. Common: vm.swappiness, net.core.somaxconn, net.ipv4.ip_local_port_range, fs.file-max, net.ipv4.tcp_tw_reuse, vm.max_map_count (Elasticsearch).

INTERVIEW TIP

Name 2–3 you actually changed and why — generic lists sound bookish.

POSSIBLE FOLLOW-UP

What does net.core.somaxconn control and when would you raise it?

Q128. Explain software RAID levels and mdadm basics.**ADVANCED****SHORT ANSWER**

RAID 0 stripes (speed, no redundancy), RAID 1 mirrors, RAID 5 stripes+parity (1-disk tolerance), RAID 6 dual parity, RAID 10 mirrored stripes (best DB choice). mdadm --create builds arrays; watch /proc/mdstat; mdadm --detail /dev/md0 inspects; failed disks are replaced with --fail/--remove/--add.

INTERVIEW TIP

cat /proc/mdstat as the first health check — say it instinctively.

POSSIBLE FOLLOW-UP

A disk in RAID 5 failed — what are your exact steps?

Q129. How do LVM snapshots work?**ADVANCED****SHORT ANSWER**

A snapshot LV records copy-on-write copies of blocks changed on the origin after creation, presenting a frozen point-in-time view — ideal for consistent backups or pre-change safety. If the snapshot fills up, it becomes invalid, so size it for expected churn.

INTERVIEW TIP

Use case: snapshot before a risky upgrade, lvconvert --merge to roll back.

POSSIBLE FOLLOW-UP

What happens when an LVM snapshot runs out of space?

Q130. What is the difference between ext4 and XFS?**ADVANCED****SHORT ANSWER**

Both are journaling filesystems. XFS (RHEL default) excels at large files, parallel I/O, and huge filesystems but cannot shrink. ext4 is mature, can shrink offline, and is common on Debian/Ubuntu. Both support online growth.

INTERVIEW TIP

'XFS cannot shrink' is the single fact most often probed.

POSSIBLE FOLLOW-UP

Which would you choose for a large database volume and why?

Q131. What is filesystem journaling?**ADVANCED****SHORT ANSWER**

The journal logs metadata (optionally data) changes before committing them, so after a crash the filesystem replays/discards incomplete transactions instead of needing a full fsck — fast, consistent recovery.

INTERVIEW TIP

Mention mount options like `data=ordered` (ext4 default).

POSSIBLE FOLLOW-UP

What is the difference between `data=ordered` and `data=journal`?

Q132. Why does a filesystem go read-only, and how do you recover?**ADVANCED****SHORT ANSWER**

The kernel remounts a filesystem read-only on I/O or metadata errors (errors=remount-ro) to prevent corruption. Check dmesg for the underlying disk/controller errors, fix or replace storage, fsck/xfs_repair from rescue if needed, then remount.

INTERVIEW TIP

Read-only FS = symptom; dying storage = cause. Check dmesg first — say it.

POSSIBLE FOLLOW-UP

What do you check in dmesg when a filesystem flips read-only?

Q133. How do you rescue a system that won't boot?**ADVANCED****SHORT ANSWER**

From GRUB, boot rescue/emergency target or an older kernel; or boot from ISO into rescue mode, mount the root filesystem, chroot, and fix (fstab, initramfs rebuild, grub2-install, filesystem repair). Root-password reset uses `rd.break` with `/sysroot` remount.

INTERVIEW TIP

Walk the `rd.break` root-password reset flow — it's asked constantly.

POSSIBLE FOLLOW-UP

Walk me through resetting a lost root password on RHEL.

Q134. How do you reset a lost root password on a systemd distro?**ADVANCED****SHORT ANSWER**

At GRUB, edit the kernel line and add `rd.break`; at the `switch_root` prompt: `mount -o remount,rw /sysroot`; `chroot /sysroot`; `passwd root`; `touch /.autorelabel` (SELinux); exit twice to continue boot.

INTERVIEW TIP

Forgetting `/.autorelabel` on SELinux systems is the trap — mention it unprompted.

POSSIBLE FOLLOW-UP

Why is `touch /.autorelabel` required?

Q135. What is /etc/nsswitch.conf?**ADVANCED****SHORT ANSWER**

It defines lookup order for name databases: hosts (files dns), passwd/group (files sssd/ldap), etc. It explains why /etc/hosts overrides DNS and how LDAP/AD users resolve alongside local ones.

INTERVIEW TIP

Tie it to real debugging: 'getent hosts name' tests the full NSS chain.

POSSIBLE FOLLOW-UP

What does 'getent passwd user' test that 'grep passwd' doesn't?

Q136. How does Linux integrate with LDAP/Active Directory?**ADVANCED****SHORT ANSWER**

Via SSSD (with realmd for AD joins): realm join domain, authentication through Kerberos, identities via LDAP, caching for offline logins. PAM and nsswitch route auth/identity to SSSD; access controlled with realm permit or simple_allow_groups.

INTERVIEW TIP

Name SSSD + realmd; the old nslcd/winbind answers sound dated.

POSSIBLE FOLLOW-UP

Users can't log in when the AD link is down — what mitigates this?

Q137. What is PAM?**ADVANCED****SHORT ANSWER**

Pluggable Authentication Modules provide a stack of auth, account, password, and session modules per service in /etc/pam.d/. It enables policies like password quality (pam_pwquality), lockouts (pam_faillock), and MFA without changing applications.

INTERVIEW TIP

Warn: a broken PAM config can lock everyone out — always keep a root session open.

POSSIBLE FOLLOW-UP

How would you implement account lockout after 5 failed logins?

Q138. How do you audit who did what on a Linux server?**ADVANCED****SHORT ANSWER**

auditd records syscall-level events per rules (auditctl, /etc/audit/rules.d); ausearch/aureport query them. Also: /var/log/secure (auth), sudo logs, bash history with timestamps, and last/lastlog for logins.

INTERVIEW TIP

auditd file-watch example: -w /etc/passwd -p wa — quote one rule.

POSSIBLE FOLLOW-UP

How do you track all modifications to /etc/sudoers?

Q139. What are Linux capabilities?**ADVANCED****SHORT ANSWER**

Capabilities split root's power into fine-grained units (e.g., CAP_NET_BIND_SERVICE to bind ports <1024, CAP_SYS_ADMIN). setcap grants them to binaries so services need not run as root; getcap inspects them.

INTERVIEW TIP

Example: setcap 'cap_net_bind_service=+ep' lets a non-root app bind port 80.

POSSIBLE FOLLOW-UP

How can a non-root service listen on port 443?

Q140. How does the kernel handle memory overcommit?**ADVANCED****SHORT ANSWER**

vm.overcommit_memory: 0 heuristic (default), 1 always allow, 2 strict (bounded by swap + ratio of RAM). Overcommit lets fork-heavy apps run but risks OOM kills later; databases like Redis document specific recommended settings.

INTERVIEW TIP

Citing Redis's overcommit_memory=1 recommendation shows real exposure.

POSSIBLE FOLLOW-UP

What is the trade-off of vm.overcommit_memory=2?

Q141. What is NUMA and why does it matter?**ADVANCED****SHORT ANSWER**

On multi-socket servers, memory attached to each CPU is faster locally than remotely (Non-Uniform Memory Access). Poor NUMA placement causes latency; numactl and numastat manage/inspect it. Databases often pin memory or use interleave.

INTERVIEW TIP

MySQL swap-storm on NUMA machines is the classic anecdote.

POSSIBLE FOLLOW-UP

How do you run a process bound to one NUMA node?

Q142. What is an I/O scheduler and which are common?**ADVANCED****SHORT ANSWER**

The I/O scheduler orders block requests: mq-deadline, kyber, bfq, and none (for NVMe where the device handles ordering). View/change via /sys/block/sdX/queue/scheduler. NVMe/SSD generally uses none; bfq favors desktops.

INTERVIEW TIP

'none for NVMe' is the modern answer — elevator/cfq answers sound outdated.

POSSIBLE FOLLOW-UP

Which scheduler would you use for NVMe and why?

Q143. How do you trace why a service is slow to start?**ADVANCED****SHORT ANSWER**

systemd-analyze blame and critical-chain reveal slow units; journalctl -u unit shows what it waited on (network, mounts, DNS). Common causes: waiting on network-online.target, slow DNS, NFS mounts, entropy (older systems).

INTERVIEW TIP

systemd-analyze critical-chain is a lesser-known gem — mention it.

POSSIBLE FOLLOW-UP

What does After=network-online.target actually guarantee?

Q144. What is the difference between soft and hard ulimits?**ADVANCED****SHORT ANSWER**

The hard limit is the ceiling set by root; the soft limit is the active value a user/process can raise up to the hard limit (ulimit -Sn / -Hn). Persisted in limits.conf; systemd services use Limit* directives instead.

INTERVIEW TIP

Repeat the systemd exception — it's tested twice in interviews for a reason.

POSSIBLE FOLLOW-UP

Can a non-root user increase their hard limit?

Q145. How do you secure/harden a Linux server?**ADVANCED****SHORT ANSWER**

Minimize packages and services; keys-only SSH with root login disabled; firewall default-deny; SELinux enforcing; automatic security updates/patching cadence; auditd; strong password/lockout policy via PAM; file integrity (AIDE); centralized logs; least-privilege sudo.

INTERVIEW TIP

Structure the answer: access, network, kernel/MAC, patching, auditing.

POSSIBLE FOLLOW-UP

Which CIS benchmark controls have you actually implemented?

Q146. What is AppArmor and how does it differ from SELinux?**ADVANCED****SHORT ANSWER**

Both are mandatory access control LSMs. AppArmor (Ubuntu/SUSE) confines programs via path-based profiles — simpler to manage. SELinux (RHEL) uses labels/contexts on every object — more granular and stricter. aa-status vs sestatus to check.

INTERVIEW TIP

Path-based vs label-based is the one-line contrast to lead with.

POSSIBLE FOLLOW-UP

What happens to AppArmor rules if a confined binary is moved?

Q147. How does SSL/TLS verification work, and how do you debug it with OpenSSL?**ADVANCED****SHORT ANSWER**

The server presents a certificate chain; the client validates signatures up to a trusted CA, plus hostname (SAN) and expiry. Debug: `openssl s_client -connect host:443 -servername host`; inspect certs with `openssl x509 -text`; check expiry dates and chain completeness.

INTERVIEW TIP

'Missing intermediate cert' as the most common real failure — name it.

POSSIBLE FOLLOW-UP

Curl works with `-k` but fails without it — what are the likely causes?

Q148. What is GPG used for on servers?**ADVANCED****SHORT ANSWER**

GPG provides asymmetric encryption and signing: verifying package/repo signatures, encrypting backups/secrets (`gpg -e -r keyid file`), and signing artifacts. Keys managed with `gpg --gen-key`, exported/imported for sharing.

INTERVIEW TIP

Tie it to package security: rpm/apt GPG verification is GPG in daily action.

POSSIBLE FOLLOW-UP

How does apt/dnf use GPG under the hood?

Q149. How do you capture and analyze a core dump?**ADVANCED****SHORT ANSWER**

Enable with `ulimit -c unlimited` (or `LimitCORE=` in `systemd`) and `kernel.core_pattern` (often `systemd-coredump`; view via `coredumpctl`). Analyze with `gdb` binary core → `bt` for the crash backtrace; forward to developers with symbols.

INTERVIEW TIP

`coredumpctl list` / `coredumpctl gdb` is the modern workflow to cite.

POSSIBLE FOLLOW-UP

Where do core dumps go when `core_pattern` is set to `systemd-coredump`?

Q150. What is chrony/NTP and why is time sync critical?**ADVANCED****SHORT ANSWER**

`chronyd` synchronizes system clocks against NTP sources (`chronyc` sources/tracking). Drifted clocks break TLS validation, Kerberos/AD auth, cluster quorum, log correlation, and certificate checks. Config in `/etc/chrony.conf`.

INTERVIEW TIP

Kerberos's 5-minute skew limit is the concrete consequence to quote.

POSSIBLE FOLLOW-UP

AD logins are failing and the clock is 10 minutes off — connect the dots.

SECTION 4

Expert Linux Questions

Kernel-level depth and SRE-grade reasoning

Q151. Walk through what happens from typing a command to seeing output.

EXPERT

SHORT ANSWER

The shell parses the line, resolves aliases/functions, searches PATH, then fork(); the child exec()s the binary, the kernel loads it via the ELF loader with dynamic linker mapping shared libraries, the process runs making syscalls, writes to stdout (fd 1) which the terminal renders; on exit the shell reaps status via wait().

INTERVIEW TIP

Structure: *parse → fork → exec → run → wait. Deliver it as five beats.*

POSSIBLE FOLLOW-UP

What exactly does the dynamic linker (ld.so) do?

Q152. Explain virtual memory and page faults.

EXPERT

SHORT ANSWER

Each process sees a private virtual address space mapped to physical frames via page tables (helped by the TLB). A minor fault maps a page already in RAM (e.g., first touch, COW); a major fault requires disk I/O (swap/file read). High major faults indicate memory pressure.

INTERVIEW TIP

Minor vs major fault distinction is exactly what separates expert answers.

POSSIBLE FOLLOW-UP

How do you see a process's major/minor fault counts?

Q153. How does copy-on-write work in fork()?

EXPERT

SHORT ANSWER

fork() duplicates page tables, marking pages read-only and shared. Writes trigger a fault; the kernel then copies just that page for the writer. This makes fork cheap regardless of process size and underlies fast redis BGSAVE-style snapshots.

INTERVIEW TIP

The redis BGSAVE example turns theory into practice.

POSSIBLE FOLLOW-UP

Why can a fork-based snapshot still cause memory spikes?

Q154. Explain the Linux CFS scheduler at a high level.

EXPERT

SHORT ANSWER

CFS gives each runnable task a fair share of CPU using virtual runtime (vruntime): the task with the smallest vruntime runs next, tracked in a red-black tree. Nice values weight vruntime accrual; cgroup CPU shares extend fairness to groups. (Recent kernels introduced EEVDF as its successor.)

INTERVIEW TIP

Say 'fairness by vruntime' — that phrase alone signals depth.

POSSIBLE FOLLOW-UP

How do cgroup cpu.weight and nice interact?

Q155. What happens exactly during the boot after the kernel loads?**EXPERT****SHORT ANSWER**

Kernel decompresses, initializes memory/SMP/drivers, mounts initramfs as rootfs and runs its init; initramfs assembles storage (LVM/RAID/crypto), mounts the real root, switch_root to it; systemd starts, reads default.target, builds the dependency graph, activates mounts/sockets/services in parallel until the target is reached.

INTERVIEW TIP

Include switch_root and the dependency graph — most candidates stop at 'systemd starts'.

POSSIBLE FOLLOW-UP

What does systemd socket activation change about service startup order?

Q156. What is systemd socket activation?**EXPERT****SHORT ANSWER**

systemd listens on a service's socket and starts the service on first connection, buffering it. Benefits: on-demand startup, parallel boot (dependents only need the socket), zero-downtime restarts since the socket persists across service restarts.

INTERVIEW TIP

Zero-downtime restart via retained socket is the expert-level benefit to name.

POSSIBLE FOLLOW-UP

How does socket activation help during a service crash?

Q157. How do you debug a process stuck in D state?**EXPERT****SHORT ANSWER**

D (uninterruptible sleep) means waiting on I/O, usually storage/NFS. It ignores signals — even kill -9. Inspect /proc/PID/stack or wchan (ps -o pid,stat,wchan) to see the kernel wait point; check dmesg for hung-task warnings, and the underlying storage/NFS server. Recovery may require fixing storage or rebooting.

INTERVIEW TIP

'kill -9 cannot kill D state' — stating that upfront wins the question.

POSSIBLE FOLLOW-UP

Why can't SIGKILL terminate a D-state process?

Q158. Explain dirty pages and writeback tuning.**EXPERT****SHORT ANSWER**

Writes land in page cache as dirty pages, flushed asynchronously by writeback threads. vm.dirty_background_ratio starts background flushing; vm.dirty_ratio blocks writers when exceeded. Tuning trades throughput vs burst-stall risk and data-loss window; fsync bypasses by forcing durability.

INTERVIEW TIP

Explain the stall symptom: giant dirty buildup → multi-second write freezes.

POSSIBLE FOLLOW-UP

Why can lowering dirty_ratio improve latency consistency?

Q159. How would you find a memory leak on a server without touching the app?**EXPERT****SHORT ANSWER**

Trend per-process RSS over time (`ps/sar/pidstat -r`); confirm it's the app not cache (free, `/proc/meminfo`). Inspect `/proc/PID/smmaps` growth areas, use `pmap` deltas; for native leaks attach `ltrace/valgrind` in staging or use eBPF-based tools (memleak). Mitigate with restart policy + MemoryMax while root-causing.

INTERVIEW TIP

The answer is a methodology: trend → localize → confirm → mitigate. Present it that way.

POSSIBLE FOLLOW-UP

How does MemoryMax in systemd behave when exceeded?

Q160. What is eBPF and how is it used in observability?**EXPERT****SHORT ANSWER**

eBPF runs sandboxed, verified programs inside the kernel attached to syscalls, kprobes, and tracepoints — enabling low-overhead tracing without kernel modules. Tools: `bcc/bpftrace` (execsnoop, opensnoop, biol latency, tcpconnect) for on-CPU, I/O latency, and network insights.

INTERVIEW TIP

Name bpftrace one-liners — e.g., execsnoop to catch what keeps spawning.

POSSIBLE FOLLOW-UP

How would you find every short-lived process spawning on a host?

Q161. Explain TCP backlog and SYN queue tuning.**EXPERT****SHORT ANSWER**

Incoming connections pass a SYN queue (`net.ipv4.tcp_max_syn_backlog`) then an accept queue bounded by `min(listen backlog, net.core.somaxconn)`. Overflow drops connections ('possible SYN flooding' or silent client timeouts). Fix: raise `somaxconn` and app backlog; SYN cookies mitigate floods.

INTERVIEW TIP

'ss -lnt shows Recv-Q vs Send-Q on listeners' — knowing where to see overflow is key.

POSSIBLE FOLLOW-UP

How do you observe accept-queue overflow in real time?

Q162. How does conntrack affect a busy server?**EXPERT****SHORT ANSWER**

Netfilter's connection tracking records every flow for stateful firewalling/NAT. The table (`net.netfilter.nf_conntrack_max`) can fill on high-traffic hosts, dropping new connections ('nf_conntrack: table full'). Fixes: raise the limit, lower timeouts, or bypass tracking (NOTRACK) for trusted high-volume flows.

INTERVIEW TIP

The `dmesg 'table full, dropping packet'` message is the smoking gun to cite.

POSSIBLE FOLLOW-UP

What symptoms indicate a full conntrack table?

Q163. What is the difference between buffered and direct I/O?**EXPERT****SHORT ANSWER**

Buffered I/O goes through page cache (fast, but double-buffering for databases with their own cache). Direct I/O (`O_DIRECT`) bypasses cache to storage for predictability and to avoid double caching; databases use it with their own buffer pools. `fsync/O_SYNC` control durability separately.

INTERVIEW TIP

Explain why databases choose `O_DIRECT` — double-caching waste and control.

POSSIBLE FOLLOW-UP

What is the difference between `O_DIRECT` and `O_SYNC`?

Q164. How does mmap-based file I/O differ from read/write?**EXPERT****SHORT ANSWER**

`mmap` maps file pages into the address space; access is via memory with faults pulling pages, no syscall per access and zero copies to user buffers. Great for random reads/shared read-mostly data; but page-cache writeback timing, TLB pressure, and `SIGBUS` on truncation complicate it.

INTERVIEW TIP

Mention `SIGBUS-on-truncate` — a genuinely expert detail.

POSSIBLE FOLLOW-UP

When would `read()` outperform `mmap`?

Q165. Explain how a zero-downtime service restart can be achieved.**EXPERT****SHORT ANSWER**

Options: socket activation (`systemd` retains the socket), `SO_REUSEPORT` with new workers before old ones drain, master-worker reload (nginx: re-exec master, graceful worker shutdown), or fronting with a load balancer draining the node. Key: never close the listening socket window.

INTERVIEW TIP

Explaining nginx's binary upgrade (`USR2/WINCH`) shows real depth.

POSSIBLE FOLLOW-UP

How does nginx reload config without dropping connections?

Q166. What are real-time scheduling policies in Linux?**EXPERT****SHORT ANSWER**

`SCHED_FIFO` and `SCHED_RR` are fixed-priority real-time policies preempting normal (`SCHED_OTHER`) tasks; `chrt` sets them. Runaway RT tasks can starve the system — bounded by `sched_rt_runtime_us`. Used for audio, telecom, and latency-critical loops.

INTERVIEW TIP

Warn about starvation and the RT throttling safeguard — balanced expertise.

POSSIBLE FOLLOW-UP

What prevents a `SCHED_FIFO` task from freezing the machine?

Q167. How do you approach CPU profiling on production?**EXPERT****SHORT ANSWER**

Start with `perf top` / `perf record -g` for on-CPU stacks, visualize as flame graphs; check context switches (`pidstat -w`), run-queue latency (`perf sched, runqlat eBPF`). Distinguish user vs sys time — high sys points to syscalls/locking; confirm with `strace -c` or off-CPU analysis.

INTERVIEW TIP

Say 'flame graph' and 'on-CPU vs off-CPU' — vocabulary of real performance work.

POSSIBLE FOLLOW-UP

What does high system (sy) CPU with low user time suggest?

Q168. Explain the VFS layer.**EXPERT****SHORT ANSWER**

The Virtual File System is the kernel abstraction (superblocks, inodes, dentries, file objects) letting syscalls work uniformly across ext4, XFS, NFS, procfs, etc. The dentry cache accelerates path lookup; filesystems implement VFS operation tables underneath.

INTERVIEW TIP

Mention dentry/inode caches — visible in `slabtop` during metadata-heavy loads.

POSSIBLE FOLLOW-UP

Why can heavy 'find' workloads consume kernel memory?

Q169. What happens when a system runs out of inodes?**EXPERT****SHORT ANSWER**

Writes fail with 'No space left on device' while `df -h` shows free space; `df -i` reveals 100% inode use — typically millions of tiny files (sessions, cache, mail queue). Fix: delete/archive small files; long-term: reformat with higher inode ratio or use a different layout/filesystem.

INTERVIEW TIP

The `df` vs `df -i` contradiction is the recognition test — answer it instantly.

POSSIBLE FOLLOW-UP

Which workloads typically exhaust inodes?

Q170. How does Linux handle swap decisions (swappiness) internally?**EXPERT****SHORT ANSWER**

The kernel balances reclaiming page cache vs swapping anonymous pages; `vm.swappiness` (0–200 in modern kernels) biases that choice. Low values prefer dropping cache; high values swap idle anon memory to keep cache hot. With cgroup v2, `memory.swap.max` controls per-service swap.

INTERVIEW TIP

Note `swappiness=0` doesn't fully disable swap under pressure — common misconception.

POSSIBLE FOLLOW-UP

Does `vm.swappiness=0` guarantee no swapping?

Q171. Explain how DNS caching works on modern Linux.**EXPERT****SHORT ANSWER**

systemd-resolved (or nscd/dnsmasq/unbound) caches lookups locally; /etc/resolv.conf may point to the 127.0.0.53 stub. Apps also cache (JVM!). Debug layers separately: resolvectl query/statistics, dig @real-server, and check app-level TTL behavior.

INTERVIEW TIP

The JVM's infinite DNS cache with security manager is a famous production trap.

POSSIBLE FOLLOW-UP

DNS record changed but one app still hits the old IP — why?

Q172. How would you diagnose intermittent network latency between two servers?**EXPERT****SHORT ANSWER**

Baseline with ping/hping (loss, jitter) and mtr for path behavior; check NIC errors/drops (ip -s link, ethtool -S), retransmits (ss -ti, nstat), qdisc drops (tc -s qdisc), conntrack limits, CPU softirq saturation (%si), and offload/IRQ affinity; capture with tcpdump during the spikes.

INTERVIEW TIP

The layered checklist (NIC → kernel → path → app) is what's being graded.

POSSIBLE FOLLOW-UP

What do rising TCP retransmissions with clean NIC counters suggest?

Q173. What is IRQ affinity and softirq pressure?**EXPERT****SHORT ANSWER**

Hardware interrupts are steered to CPUs (smp_affinity, irqbalance); packet processing continues in softirqs (NET_RX shown as %si). One CPU saturated with %si while others idle means poor RSS/affinity spreading — fix with multi-queue NICs, RSS/RPS tuning.

INTERVIEW TIP

top showing one core at 100% si is the recognizable symptom to describe.

POSSIBLE FOLLOW-UP

One CPU core is pinned at 100% softirq — what do you do?

Q174. Explain filesystem barriers/fsync guarantees and data loss windows.**EXPERT****SHORT ANSWER**

Durability requires fsync (and the device honoring flushes). Data in page cache is lost on power failure; journals protect metadata consistency, not unsynced data. Write barriers/FUA ensure journal ordering to media; disabling volatile write caches without protection risks corruption.

INTERVIEW TIP

'Journaling ≠ zero data loss' — draw that line explicitly.

POSSIBLE FOLLOW-UP

Why can a database still lose data on a battery-less RAID cache?

Q175. How do you keep systems patched with minimal risk at scale?**EXPERT****SHORT ANSWER**

Staged rings (dev → canary → prod waves), configuration management (Ansible) for consistency, dnf-automatic/unattended-upgrades for security patches, kernel live patching (kpatch/livepatch) for critical CVEs without reboot, plus scheduled reboot windows, health checks, and rollback plans (snapshots, previous kernel retained).

INTERVIEW TIP

Mention live patching plus staged rings — that's the enterprise-grade answer.

POSSIBLE FOLLOW-UP

When is kernel live patching not sufficient?

Q176. What is an initramfs emergency you have handled (dracut shell)?**EXPERT****SHORT ANSWER**

Typical case: root LVM/UUID not found after storage or fstab changes → dracut emergency shell. Fix: inspect with lsblk/blkid, correct root= or /etc/fstab from rescue, rebuild initramfs (dracut -f) including needed drivers, regenerate GRUB config, reboot.

INTERVIEW TIP

Tell it as a story with the exact commands — scenario delivery matters at this level.

POSSIBLE FOLLOW-UP

What causes 'dracut-initqueue timeout' at boot?

Q177. How do you contain a compromised Linux server?**EXPERT****SHORT ANSWER**

Isolate at network level (SG/firewall) without powering off; preserve evidence (memory capture, disk snapshot); collect volatile data (connections, processes, users); identify persistence (cron, systemd units, SSH keys, ld.so.preload); rotate credentials; rebuild from clean images rather than 'cleaning'.

INTERVIEW TIP

'Rebuild, don't clean' and 'don't power off before capture' are the two expert markers.

POSSIBLE FOLLOW-UP

Where do attackers commonly hide persistence on Linux?

Q178. Explain ld.so.preload-based attacks and defenses.**EXPERT****SHORT ANSWER**

/etc/ld.so.preload forces libraries into every dynamic binary — rootkits use it to hook libc calls and hide processes/files. Detect by comparing static tools (busybox) vs system tools, checking the file itself, AIDE integrity baselines, and auditd watches on loader configs.

INTERVIEW TIP

The static-binary comparison trick shows genuine IR experience.

POSSIBLE FOLLOW-UP

Why might ls hide files while a statically linked ls shows them?

Q179. How does systemd manage dependencies and ordering differently?**EXPERT****SHORT ANSWER**

Requires/Wants define dependency (start together; Requires fails with the dependency), while After/Before define only ordering. A unit can Want another yet start in parallel unless After is set. Misunderstanding this causes race conditions at boot.

INTERVIEW TIP

'Wants without After = parallel start' — that sentence resolves most boot races.

POSSIBLE FOLLOW-UP

Service starts before the network is ready despite Wants=network-online.target — why?

Q180. Describe your approach to a full performance triage on an unknown slow server.**EXPERT****SHORT ANSWER**

USE method: for each resource (CPU, memory, disk, network) check Utilization, Saturation, Errors — uptime/top, vmstat, free, iostat -x, ss/nstat, dmesg. 60-second first pass, then drill into the saturated resource with perf/eBPF, correlate with recent changes and sar history.

INTERVIEW TIP

Naming the USE method and a 60-second checklist is a senior signature.

POSSIBLE FOLLOW-UP

Which 5 commands do you run in the first minute and why?

SECTION 5

Real MNC Linux Questions

Question patterns reported from top product and services companies

Q181. [Google] A server's load average is 80 on 16 cores but CPU is mostly idle. Diagnose.

ADVANCED**SHORT ANSWER**

Load counts runnable plus D-state (uninterruptible) tasks. Idle CPU with huge load means processes blocked on I/O — check vmstat's b column, iostat -x for saturated disks, ps -eo stat | grep D, and NFS/storage health via dmesg.

INTERVIEW TIP

Immediately separate CPU-bound vs I/O-bound load — that split IS the answer.

POSSIBLE FOLLOW-UP

How do you list all D-state processes with their kernel wait channel?

Q182. [Amazon] An EC2 Linux instance is unreachable over SSH. What is your systematic approach?

ADVANCED**SHORT ANSWER**

Check instance status checks and console screenshot/logs; verify security group and NACL allow 22; confirm the instance's public IP/route; use SSM Session Manager or serial console to get in; then inspect sshd status, host firewall, fstab-induced emergency mode, or full disk.

INTERVIEW TIP

The console screenshot + SSM fallback shows AWS operational maturity.

POSSIBLE FOLLOW-UP

The console shows the boot stuck in emergency mode — what next?

Q183. [Microsoft] Explain the difference between a process being 'stopped' and 'sleeping'.

INTERMEDIATE**SHORT ANSWER**

Sleeping (S/D) means waiting for an event or I/O — normal states. Stopped (T) means execution suspended by SIGSTOP/SIGTSTP (e.g., Ctrl+Z or a debugger) and resumable with SIGCONT. D-state sleep cannot be interrupted by signals.

INTERVIEW TIP

Map states to ps STAT letters: R, S, D, T, Z — recite them.

POSSIBLE FOLLOW-UP

What does the Z state indicate and how is it cleared?

Q184. [Meta] How would you find every process that has a specific deleted library still mapped?**EXPERT****SHORT ANSWER**

After patching (e.g., OpenSSL), processes keep old mappings: `lsuf | grep 'DEL.*libssl'` or `grep -l 'libssl.*deleted' /proc/*/maps` identify them; on RHEL, 'dnf needs-restarting' lists services requiring restart. Restart those services to load the fixed library.

INTERVIEW TIP

This is the 'why patching isn't done until restart' question — say that explicitly.

POSSIBLE FOLLOW-UP

Why isn't a library patch effective until services restart?

Q185. [Netflix] CPU usage is low, memory is fine, but request latency spiked. Where do you look?**EXPERT****SHORT ANSWER**

Look beyond averages: run-queue latency (`runqlat`), disk latency percentiles (`biolatency`), TCP retransmits (`ss -ti, nstat`), `conntrack/accept-queue` overflows, GC pauses in the app, DNS slowness, and noisy neighbors/CPU throttling (`cgroup cpu.stat throttled time`).

INTERVIEW TIP

Netflix-style answer = percentiles and eBPF tools, not top. Mention throttling in cgroups.

POSSIBLE FOLLOW-UP

How do you detect cgroup CPU throttling for a service?

Q186. [Oracle] A database team reports huge pages are not being used. How do you verify and fix?**ADVANCED****SHORT ANSWER**

Check `/proc/meminfo` (`HugePages_Total/Free`) and the DB's startup logs; ensure `vm.nr_hugepages` is set (and `memlock ulimits` for the DB user), pages reserved before fragmentation (set at boot), and THP disabled per vendor guidance.

INTERVIEW TIP

Mention allocating huge pages at boot to avoid fragmentation failures.

POSSIBLE FOLLOW-UP

Why can setting `nr_hugepages` at runtime fail on a long-running server?

Q187. [IBM] Explain how you would migrate a physical server's storage to LVM with minimal downtime.**ADVANCED****SHORT ANSWER**

Add new disk(s), `pvccreate` and build a VG/LV layout, `rsync` data while live, do a short cutover `rsync` with the app briefly stopped, switch `mounts/fstab`, keep old disk as rollback. For root, use `pvmove`-based approaches or rebuild via rescue window.

INTERVIEW TIP

Emphasize the two-pass `rsync` cutover pattern — the standard low-downtime trick.

POSSIBLE FOLLOW-UP

What does `pvmove` do and is it safe online?

Q188. [Cisco] The NIC shows increasing RX drops. What are the possible layers of cause?**ADVANCED****SHORT ANSWER**

Check `ethtool -S` for ring buffer overruns (raise with `ethtool -G`), CPU softirq saturation (`%si`, one hot core → fix RSS/IRQ affinity), backlog (`netdev_max_backlog`), and application slow reads causing socket buffer overflow (`ss -m`). Also physical/duplex errors.

INTERVIEW TIP

Order it NIC ring → kernel backlog → socket → app: layer-by-layer thinking.

POSSIBLE FOLLOW-UP

What does `ethtool -G` change and when do you use it?

Q189. [Red Hat] What happens when you run `systemctl enable` on a unit, exactly?**INTERMEDIATE****SHORT ANSWER**

It creates symlinks per the unit's [Install] section — typically in `/etc/systemd/system/multi-user.target.wants/` pointing to the unit file — so the target pulls the unit at boot. It does not start the unit now; `disable` removes the symlinks.

INTERVIEW TIP

Answering 'it creates symlinks per [Install]' precisely is what Red Hat wants.

POSSIBLE FOLLOW-UP

What is the difference between `enable`, `enable --now`, and `mask`?

Q190. [VMware] A VM's clock drifts badly. What do you check on the Linux guest?**INTERMEDIATE****SHORT ANSWER**

Verify `chronyd` is running and sources are reachable (`chronyc tracking/sources`); check the clocksource (`/sys/devices/system/clocksource/.../current_clocksource` — `kvm-clock/tsc` for VMs); avoid double-sync (hypervisor tools + NTP both stepping time).

INTERVIEW TIP

Mention the double time-sync conflict — the classic virtualization pitfall.

POSSIBLE FOLLOW-UP

Why is periodic hypervisor time-sync plus NTP a problem?

Q191. [Adobe] Users report 'Permission denied' on a share although POSIX permissions look correct. What else can deny access?**ADVANCED****SHORT ANSWER**

Parent-directory execute bits missing, ACLs (`getfacl`), SELinux contexts (`ls -Z`, `audit.log`), immutable attribute (`lsattr`), mount options (`ro`, `noexec`), NFS `root_squash`/ID mismatch, or AppArmor confinement.

INTERVIEW TIP

List 4+ non-obvious layers quickly — breadth is the test here.

POSSIBLE FOLLOW-UP

How does `root_squash` cause root to get permission denied on NFS?

Q192. [ServiceNow] How do you roll out a config change to 500 Linux servers safely?**ADVANCED****SHORT ANSWER**

Use configuration management (Ansible) with the change in version control; lint/dry-run (--check --diff), canary on a small batch, serial rolling with health checks and max-fail thresholds, and instant rollback via the previous git revision.

INTERVIEW TIP

Canary + serial + rollback is the expected vocabulary trio.

POSSIBLE FOLLOW-UP

How do you prevent configuration drift between servers?

Q193. [Accenture] A client's app user must run exactly two privileged commands. Design the sudo access.**INTERMEDIATE****SHORT ANSWER**

In `/etc/sudoers.d/` (edited with `visudo -f`): `'appuser ALL=(root) NOPASSWD: /usr/bin/systemctl restart app, /usr/sbin/tcpdump'`. Use absolute paths, no wildcards that expand scope, and log via sudo's default logging.

INTERVIEW TIP

Warn about wildcard sudo rules enabling privilege escalation.

POSSIBLE FOLLOW-UP

Why is `'appuser ALL=(root) /bin/bash'` or wildcard usage dangerous?

Q194. [Deloitte] Audit finding: world-writable files must be eliminated. How do you find and fix them?**INTERMEDIATE****SHORT ANSWER**

`find / -xdev -type f -perm -0002 -not -path '/proc/*'` lists world-writable files; review before `chmod o-w`. Also `find -perm -0002 -type d ! -perm -1000` for dirs missing sticky bit. Automate a recurring compliance scan.

INTERVIEW TIP

Mention reviewing before mass-chmod — blind fixes break apps.

POSSIBLE FOLLOW-UP

Why is the sticky bit acceptable on world-writable directories?

Q195. [Capgemini] Explain the complete lifecycle of a log line from an app to a central server.**INTERMEDIATE****SHORT ANSWER**

App writes to stdout/file or syslog → journald/rsyslog captures with metadata → local rules filter/route → forwarded via RELP/TCP (rsyslog) or shippers (Filebeat/Fluentd) to central store (ELK/Loki) → index, retention, and alerting policies apply.

INTERVIEW TIP

Show you know both classic rsyslog forwarding and modern shippers.

POSSIBLE FOLLOW-UP

How do you forward only auth failures to a SIEM using rsyslog?

Q196. [Infosys] What is the difference between soft mount and hard mount in NFS, and which do you choose for a database?**ADVANCED****SHORT ANSWER**

Hard mounts retry forever (apps block until the server returns) preserving data integrity; soft mounts time out and return I/O errors, risking corruption for writes. For databases: hard,intr-style with proper timeouts — never soft for write-critical data.

INTERVIEW TIP

Take a firm stance: 'soft mounts risk silent corruption for writes'.

POSSIBLE FOLLOW-UP

What does the intr/timeo/retrans tuning do on NFS mounts?

Q197. [TCS] A cron job runs fine manually but fails in cron. List all likely causes.**INTERMEDIATE****SHORT ANSWER**

Minimal PATH/environment in cron, relative paths, missing DISPLAY/tty, different shell (sh vs bash), permissions of the script, environment variables loaded only in interactive profiles, output not captured hiding errors, or crontab syntax/percent-sign issues.

INTERVIEW TIP

Name 5 causes rapidly — this question tests checklist recall.

POSSIBLE FOLLOW-UP

How does % behave inside a crontab command line?

Q198. [Wipro] How do you check if a server was rebooted and why?**INTERMEDIATE****SHORT ANSWER**

uptime/who -b for last boot time; last reboot for history; journalctl -b -1 -e for the previous boot's final messages; /var/log/messages and dmesg for panic/OOM traces; check wtmp and any power events; audit logs for shutdown commands and the initiating user.

INTERVIEW TIP

journalctl -b -1 (previous boot) is the exact flag they hope you know.

POSSIBLE FOLLOW-UP

How do you see logs from the boot before the crash?

Q199. [HCL] What steps do you take before and after patching a production RHEL server?**INTERMEDIATE****SHORT ANSWER**

Before: change ticket, snapshot/backup, note kernel/pkg versions (uname -r, dnf history), check disk space in /boot, review errata. Patch: dnf update in window. After: reboot if kernel/glibc changed, verify services and app health, dnf needs-restarting, document; rollback = previous kernel or dnf history undo.

INTERVIEW TIP

Mentioning /boot space and needs-restarting shows real patching scars.

POSSIBLE FOLLOW-UP

When is a reboot mandatory after patching?

Q200. [Cognizant] Explain the difference between systemctl mask and disable.**BASIC****SHORT ANSWER**

disable removes boot-time symlinks but the unit can still be started manually or by dependencies. mask links the unit to /dev/null, making any start attempt fail — a hard block until unmasked.

INTERVIEW TIP

Use case: masking a conflicting service during migration.

POSSIBLE FOLLOW-UP

When would you mask a service in production?

Q201. [EPAM] You must transfer 2TB between data centers nightly. Design the approach.**ADVANCED****SHORT ANSWER**

rsync -aH --delete over SSH with compression tuned (or none on fast links), parallelize by top-level directories, use --partial for resumability, schedule via systemd timer with locking (flock), monitor duration/exit codes, and consider snapshots for a consistent source.

INTERVIEW TIP

flock to prevent overlapping runs is the production-grade detail.

POSSIBLE FOLLOW-UP

How do you prevent two overlapping rsync runs of the same job?

Q202. [ThoughtWorks] How do you make a bash script safe for production?**INTERMEDIATE****SHORT ANSWER**

set -euo pipefail; quote all variables; use mktemp for temp files; trap cleanup on EXIT; check dependencies; explicit absolute paths; meaningful exit codes and logging; shellcheck in CI; idempotent operations and a dry-run flag.

INTERVIEW TIP

'set -euo pipefail plus shellcheck' is the expected opening line.

POSSIBLE FOLLOW-UP

What does pipefail change in a pipeline's exit status?

Q203. [Publicis Sapient] A JVM app on Linux shows OutOfMemory but the heap is small. Explain possibilities.**EXPERT****SHORT ANSWER**

Native (off-heap) memory: metaspace, thread stacks (thousands of threads), direct byte buffers, native libs; or the container/cgroup memory limit being hit (kernel OOM kill, exit 137), or ulimit on address space/max processes. Check cgroup memory.events, dmesg, and NativeMemoryTracking.

INTERVIEW TIP

Exit code 137 = cgroup OOM kill — connecting that earns the point.

POSSIBLE FOLLOW-UP

What does container exit code 137 signify?

Q204. [Persistent] Describe how you'd set up passwordless SSH for an automation account securely. **INTERMEDIATE****SHORT ANSWER**

Dedicated account with ssh-keygen (ed25519), public key distributed to targets' authorized_keys with from= and command= restrictions where possible, private key protected (vault/agent), PermitRootLogin no, and sudo rules limited to required commands with logging.

INTERVIEW TIP

Mention authorized_keys options (from=, command=) — few candidates do.

POSSIBLE FOLLOW-UP

How does the command= option in authorized_keys work?

Q205. [Google] How do you limit blast radius when running a risky command fleet-wide? **ADVANCED****SHORT ANSWER**

Canary on 1%, then staged waves with automatic health verification and abort thresholds; dry-run/diff first; timeouts and rate limits; feature-flag style toggles; per-host locks; and instant rollback artifacts. Never 100% in one shot.

INTERVIEW TIP

Percent-based staged rollout language is exactly what SRE interviewers want.

POSSIBLE FOLLOW-UP

What automated signal would abort your rollout?

Q206. [Amazon] Disk space on / is 100% and you cannot even create temp files. Recover it live. **ADVANCED****SHORT ANSWER**

Find quick wins: truncate huge logs (: > file, not rm on open files), clear package caches, journalctl --vacuum-size, check /tmp and core dumps, du -xh --max-depth=1 / to localize, lsof +L1 for deleted-but-open space, and move data off if needed. Root-cause with growth trend afterward.

INTERVIEW TIP

Truncate-not-delete for open logs is the key production reflex.

POSSIBLE FOLLOW-UP

Why does deleting an open log file not free space?

Q207. [Microsoft] How does Linux decide which DNS server and search domain to use with systemd-resolved? **ADVANCED****SHORT ANSWER**

resolved builds per-link DNS config (from DHCP/NM), with global fallbacks; /etc/resolv.conf usually points to the 127.0.0.53 stub. Queries route by link and search domains (~. for default). Inspect with resolvectl status/query; per-link domains explain split-DNS/VPN behavior.

INTERVIEW TIP

Split-DNS via per-link routing domains is the modern gotcha to mention.

POSSIBLE FOLLOW-UP

After connecting to VPN, only some domains resolve — why?

Q208. [Meta] What is the difference between throughput and latency tuning on Linux, with one example each?**EXPERT****SHORT ANSWER**

Throughput tuning maximizes work/second — bigger buffers, batching, deeper queues (e.g., raising net buffer sizes, NIC ring, dirty ratios). Latency tuning minimizes response time — smaller queues, fewer coalesced interrupts, disabling deep C-states, low dirty thresholds. They often conflict; tune per SLO.

INTERVIEW TIP

State the conflict explicitly and anchor to SLOs — systems-thinking answer.

POSSIBLE FOLLOW-UP

How does interrupt coalescing trade latency for throughput?

Q209. [Red Hat] How do you build a custom systemd service with resource limits and hardening?**ADVANCED****SHORT ANSWER**

Unit with [Service]: ExecStart, User/Group non-root, Restart=on-failure, MemoryMax/CPUQuota, LimitNOFILE; hardening: ProtectSystem=strict, ProtectHome, PrivateTmp, NoNewPrivileges, CapabilityBoundingSet minimal; then daemon-reload, enable --now, verify with systemd-analyze security.

INTERVIEW TIP

systemd-analyze security scoring the unit is a standout mention.

POSSIBLE FOLLOW-UP

What does PrivateTmp=yes actually do?

Q210. [Netflix] Explain how you'd debug 'connection timed out' vs 'connection refused'.**INTERMEDIATE****SHORT ANSWER**

Refused = host reachable, port closed/no listener (RST returned) → check the service and its bind address. Timed out = packets not answered → firewall drop, wrong route/SG, host down, or filtering — trace with tcpdump on both ends and check DROP vs REJECT semantics.

INTERVIEW TIP

RST vs silent drop mapping to REJECT vs DROP is the crisp core.

POSSIBLE FOLLOW-UP

Which firewall action causes each of these errors?

SECTION 6

Production Scenario Questions

Incident-style questions: problem, short solution, and how to present it

Q211. Scenario: CPU at 100%**ADVANCED****PROBLEM**

The server's CPU is pinned at 100% and the application is slow.

SHORT SOLUTION

top/htop to find the top consumers and whether it's user or system time; pidstat 1 for per-process trend; if system-heavy, check runaway syscalls (strace -c) or interrupts; mitigate by renicing/limiting (cgroup CPUQuota) or restarting the offender; root-cause via perf/flame graph.

INTERVIEW TIP

Always classify us vs sy vs wa before touching anything.

Q212. Scenario: Memory Leak**ADVANCED****PROBLEM**

A service's memory grows daily until the server degrades.

SHORT SOLUTION

Confirm with trend data (sar -r, pidstat -r): rising RSS of one process vs healthy cache growth. Mitigate: scheduled restart + systemd MemoryMax as a guardrail. Root-cause: smaps/pmap growth, heap profilers, or eBPF memleak in staging.

INTERVIEW TIP

Distinguish app leak from page cache — 'available' memory is the truth, not 'free'.

Q213. Scenario: Swap Full**ADVANCED****PROBLEM**

Swap is 100% used and performance collapsed.

SHORT SOLUTION

vmstat: continuous si/so means active thrashing. Find top swap consumers (grep VmSwap /proc/*/status | sort). Stop/restart the memory hogs, add RAM/temporary swapfile if needed, tune vm.swappiness, and fix the underlying over-allocation.

INTERVIEW TIP

Used-but-idle swap is fine; active si/so traffic is the real red flag.

Q214. Scenario: Filesystem Full**ADVANCED****PROBLEM**

df shows / at 100%; applications are failing to write.

SHORT SOLUTION

du -xh --max-depth=1 / to localize; typical culprits: /var/log, caches, core dumps. Truncate open logs (: > log), vacuum the journal, clear package cache; verify with lsof +L1 for deleted-open files; then enforce logrotate and disk alerts.

INTERVIEW TIP

Never rm an open log to free space — truncate it.

Q215. Scenario: Inode Full**ADVANCED****PROBLEM**

Disk shows free space, but writes fail with 'No space left on device'.

SHORT SOLUTION

`df -i` confirms inode exhaustion. Locate small-file explosions: `for d in /*; do echo $d $(find $d -xdev | wc -l); done` — often sessions, mail queue, or cache dirs. Delete/archive them; long-term, restructure or `mkfs` with more inodes.

INTERVIEW TIP

The free-space-but-no-space contradiction should trigger 'df -i' instantly.

Q216. Scenario: SSH Not Working**ADVANCED****PROBLEM**

You cannot SSH into a production server.

SHORT SOLUTION

From outside: ping, then `nc -zv host 22` to separate network vs service. Via console: `systemctl status sshd`, `journalctl -u sshd`, `sshd -t` for config errors, firewall rules, disk full (sshd can't write), and recent `sshd_config` changes; check fail2ban/DenyHosts blocks.

INTERVIEW TIP

Always test the port with nc before blaming sshd — isolate the layer first.

Q217. Scenario: DNS Not Resolving**ADVANCED****PROBLEM**

Applications fail with 'name or service not known'.

SHORT SOLUTION

Test layers: `getent hosts name` (NSS), `resolvectl query / dig name (stub)`, `dig @upstream name (server)`. Check `/etc/resolv.conf` correctness, resolved status, firewall on 53, and search-domain issues. Fix the failing layer; flush caches after.

INTERVIEW TIP

getent vs dig separation shows whether NSS config or DNS servers are at fault.

Q218. Scenario: Permission Denied**ADVANCED****PROBLEM**

A user gets 'Permission denied' though file permissions look right.

SHORT SOLUTION

Check the whole path's execute bits (`namei -l /full/path`), ACLs (`getfacl`), SELinux (`ausearch -m AVC`), immutable attr (`lsattr`), and mount options (`noexec/ro`). Fix the specific layer, e.g., restorecon for contexts.

INTERVIEW TIP

namei -l is the one-command path-permission auditor — quote it.

Q219. Scenario: SELinux Blocking**ADVANCED****PROBLEM**

A web app fails to read its files; permissions are correct.

SHORT SOLUTION

getenforce confirms Enforcing; ausearch -m AVC -ts recent shows denials; audit2why explains. Fix properly: restorecon -Rv on the content, semanage fcontext for custom paths, or setsebool -P for feature booleans. Do not disable SELinux.

INTERVIEW TIP

'Fix the context, not the mode' — never answer 'setenforce 0' as the solution.

Q220. Scenario: Firewall Blocking**ADVANCED****PROBLEM**

A newly deployed service is unreachable from clients.

SHORT SOLUTION

Confirm it listens on the right address (ss -tlnp — 127.0.0.1 vs 0.0.0.0); test locally with curl; then firewall-cmd --list-all / iptables -L -n for missing rules; add service/port permanently and reload; also verify cloud SG/NACL layers.

INTERVIEW TIP

Bind address vs firewall are the two 80% causes — check bind first.

Q221. Scenario: Service Failed**ADVANCED****PROBLEM**

systemctl shows a unit in failed state after deployment.

SHORT SOLUTION

systemctl status for the exit code, journalctl -u unit -xe for logs; classic causes: config syntax (validate with app's test flag), missing env/permissions, port in use, wrong ExecStart path. Fix, daemon-reload if the unit changed, restart, and add Restart=on-failure.

INTERVIEW TIP

Read the exit/status code aloud — 203 means exec not found, for example.

Q222. Scenario: Kernel Panic**ADVANCED****PROBLEM**

Server crashed with a kernel panic overnight.

SHORT SOLUTION

Check console/BMC screenshot and /var/crash vmcore (if kdump enabled); boot the previous kernel from GRUB if a recent update triggered it; review recently added modules/drivers, memtest hardware; open a vendor case with the vmcore.

INTERVIEW TIP

Previous-kernel boot from GRUB is the immediate service-restoration move.

Q223. Scenario: GRUB Failure**ADVANCED****PROBLEM**

The server drops to a grub rescue prompt at boot.

SHORT SOLUTION

From grub rescue: ls partitions, set prefix/root, insmod normal to boot manually; then permanently repair from the OS or rescue ISO: reinstall GRUB (grub2-install or EFI entry), regenerate grub.cfg, and check /boot contents and disk health.

INTERVIEW TIP

Knowing the manual grub-rescue boot sequence separates seniors from juniors.

Q224. Scenario: Server Not Booting**ADVANCED****PROBLEM**

After maintenance, the server hangs in emergency mode.

SHORT SOLUTION

Read the journal prompt (journalctl -xb): most common is a bad/missing fstab device. Remount / rw, fix fstab (add nofail or correct UUID via blkid), or rebuild initramfs if storage drivers changed; reboot and verify mounts.

INTERVIEW TIP

Bad fstab is the #1 cause of emergency mode — say it, then prove with journalctl -xb.

Q225. Scenario: High Load Average**ADVANCED****PROBLEM**

Load average is 4x the core count; users report slowness.

SHORT SOLUTION

Classify: top for CPU-bound (high us/sy) vs I/O-bound (high wa, D-state tasks — ps -eo stat,pid,cmd | grep '^D'). CPU-bound → profile/limit top consumers; I/O-bound → iostat -x to find the saturated device and its writer (iotop).

INTERVIEW TIP

Load without classification is meaningless — always split CPU vs I/O first.

Q226. Scenario: Zombie Process**ADVANCED****PROBLEM**

Hundreds of zombie processes are accumulating.

SHORT SOLUTION

Zombies are unreaped children — find the parent (ps -eo pid,ppid,stat | awk '\$3~/Z/'), then fix or restart that parent service; SIGCHLD to the parent sometimes triggers reaping. Zombies hold no memory, but exhaust PIDs at scale.

INTERVIEW TIP

You can't kill a zombie — restart the parent. Lead with that.

Q227. Scenario: Disk Failure**ADVANCED****PROBLEM**

SMART alerts and I/O errors are appearing for /dev/sdb.

SHORT SOLUTION

Confirm: `smartctl -a /dev/sdb`, `dmesg` I/O errors. If in RAID/LVM-mirror, fail and replace the member (`mdadm --fail/--remove`, insert, `--add`, watch rebuild in `/proc/mdstat`). If standalone, restore from backup to a new disk; never trust a failing disk.

INTERVIEW TIP

State the RAID member replacement flow in one breath — it's the expected core.

Q228. Scenario: RAID Failure**ADVANCED****PROBLEM**

A RAID 5 array is degraded after a disk died.

SHORT SOLUTION

`cat /proc/mdstat` and `mdadm --detail` to confirm which member failed; replace: `mdadm --fail/--remove /dev/md0` faulty, physically swap, partition identically, `mdadm --add`, monitor rebuild; remember RAID 5 tolerates only one loss — backup before rebuild stress.

INTERVIEW TIP

Mention rebuild-window risk: a second failure during rebuild loses the array.

Q229. Scenario: NFS Mount Failure**ADVANCED****PROBLEM**

Clients cannot mount an NFS share; existing mounts hang.

SHORT SOLUTION

Server side: `systemctl status nfs-server`, `exportfs -v`, `showmount -e localhost`. Network: `rpcinfo -p server`, firewall for 2049. Client: `mount -vvv` output, hard-mount hangs indicate server/network outage — restore server first, then remount (`umount -l` for stuck mounts).

INTERVIEW TIP

Hung `df/l`s on the client is the signature of a dead hard NFS mount.

Q230. Scenario: Read-Only Filesystem**ADVANCED****PROBLEM**

Applications crash with 'Read-only file system' errors.

SHORT SOLUTION

The kernel remounted RO on errors — `dmesg` confirms the I/O/journal error and device. Check storage health (`smartctl`, controller/SAN paths); after fixing, `fsck/xfs_repair` from rescue, then mount `rw`; investigate multipath/SAN blips for VMs.

INTERVIEW TIP

Treat RO remount as a storage incident, not a mount option mistake.

Q231. Scenario: Log Rotation Failure**ADVANCED****PROBLEM**

A log grew to 50GB; logrotate seems to do nothing.

SHORT SOLUTION

Test: `logrotate -d /etc/logrotate.d/app` for config errors; check state file timestamps, cron/timer execution, and whether the app reopens files post-rotation (needs `postrotate reload` or `copytruncate`). Fix the config and force one rotation (`-f`).

INTERVIEW TIP

copytruncate vs postrotate-reload is the decision point interviewers probe.

Q232. Scenario: Cron Not Running**ADVANCED****PROBLEM**

A scheduled job silently never runs.

SHORT SOLUTION

Verify `crond` is active; check the job's log trail (`journalctl -u crond, /var/log/cron`); confirm crontab syntax and user; test the command with cron's environment (minimal `PATH`); check permissions/executable bit and `cron.allow/deny`; capture output to a log for evidence.

INTERVIEW TIP

Add `'>> /var/log/job.log 2>&1'` to the entry — evidence beats guessing.

Q233. Scenario: Time Sync Issue**ADVANCED****PROBLEM**

Server clock drifted 6 minutes; Kerberos/AD auth is failing.

SHORT SOLUTION

`chronyc tracking/sources` to see sync state; if far off, `chronyc makestep` to step immediately; ensure NTP sources reachable (firewall UDP 123), no conflicting sync (VM tools + `chrony`), then confirm auth recovers; add drift monitoring.

INTERVIEW TIP

Link the symptom: Kerberos rejects tickets beyond ~5 minutes of skew.

Q234. Scenario: Application Not Starting**ADVANCED****PROBLEM**

A deployed app exits immediately on start.

SHORT SOLUTION

Run it in the foreground to see the real error; check `journalctl -u app`, exit code meaning, config validation, missing env vars/secrets, dependency ports/DB reachability, file permissions/SELinux, and library versions (ldd binary for missing libs).

INTERVIEW TIP

Foreground run for the raw error is step one — logs may be swallowed.

Q235. Scenario: Port Already in Use**ADVANCED****PROBLEM**

Service fails: 'bind: address already in use'.

SHORT SOLUTION

`ss -tlnp | grep :PORT` (or `lsof -i :PORT`) identifies the holder — stale old instance, another app, or systemd socket unit. Stop/kill the holder or change the port; for restarts, ensure `SO_REUSEADDR`/proper shutdown; check for a masked leftover process.

INTERVIEW TIP

Also mention systemd socket units can hold the port — a subtle catch.

Q236. Scenario: Network Failure**ADVANCED****PROBLEM**

A server lost all network connectivity after a change.

SHORT SOLUTION

Console access: `ip link` (interface up?), `ip addr` (IP present?), `ip route` (default gw?), `ping gw` → DNS test separately. Check NIC driver messages in `dmesg`, NetworkManager/netplan config errors, and roll back the last change; verify switch/VLAN side too.

INTERVIEW TIP

Debug in OSI order — link, IP, route, DNS — and say you're doing so.

Q237. Scenario: Slow Server**ADVANCED****PROBLEM**

Everything is 'slow' but no single alert fired.

SHORT SOLUTION

60-second triage: `uptime`, `dmesg -T | tail`, `vmstat 1`, `iostat -xz 1`, `free -h`, `sar -n DEV`, `top`. Identify the saturated resource (USE method), then drill down with the matching tool; compare against `sar` history and recent changes/deployments.

INTERVIEW TIP

A named methodology (USE + 60-second checklist) is the winning structure.

Q238. Scenario: Package Dependency Error**ADVANCED****PROBLEM**

`dnf update` fails with broken/conflicting dependencies.

SHORT SOLUTION

Read the exact conflict; try `dnf clean all + retry`, check for mixed/stale repos (`dnf repolist`, `.repo` files), excluded/locked versions (`versionlock`), and duplicate rpms (`package-cleanup --dupes` / `dnf remove --duplicates`); use `dnf history` to recover a half-done transaction; `--allow-erasing` only after review.

INTERVIEW TIP

Blindly using `--skip-broken`/`--allow-erasing` is a red flag — always read the conflict.

Q239. Scenario: File Descriptor Limit**ADVANCED****PROBLEM**

App logs show 'Too many open files' under load.

SHORT SOLUTION

Check the process limit: `cat /proc/PID/limits`, count usage: `ls /proc/PID/fd | wc -l`. Raise via `LimitNOFILE=` in the systemd unit (`daemon-reload + restart`) or `limits.conf` for user sessions; also verify it isn't a leak (`CLOSE_WAIT`/unclosed files) before raising.

INTERVIEW TIP

Rule out an FD leak before raising limits — raising just delays the crash.

Q240. Scenario: OOM Killer Triggered**ADVANCED****PROBLEM**

A critical process disappeared overnight; `dmesg` shows 'Out of memory: Killed'.

SHORT SOLUTION

Confirm via `dmesg/journalctl -k` the victim and scores; identify the true memory hog at that time (`sar -r`, `atop` logs); fix: right-size memory, add swap headroom, set `MemoryMax` on offenders and `OOMScoreAdjust=-1000` on critical services; consider `earlyoom/systemd-oomd`.

INTERVIEW TIP

The killed process is often the victim, not the culprit — say that line.

SECTION 7

Troubleshooting Questions

Methodology questions that reveal how you actually debug

Q241. A user says 'the server is down'. What is your first minute?**ADVANCED****SHORT ANSWER**

Verify scope: can you ping/SSH? Is it one service or the host? Check monitoring/alerts, console access if SSH fails, then uptime, dmesg -T | tail, systemctl --failed, df -h, free -h, top — establishing whether it's network, resource, or service level.

INTERVIEW TIP

Show triage discipline: verify, scope, then dig — don't jump to fixes.

POSSIBLE FOLLOW-UP

How do you decide between restoring service fast vs preserving evidence?

Q242. How do you find which process is writing heavily to disk?**ADVANCED****SHORT ANSWER**

iostat -oPa shows per-process I/O live; pidstat -d 1 gives per-process disk stats; for deeper attribution, eBPF biosnoop/biotop maps I/O to processes; correlate with iostat -x device saturation.

INTERVIEW TIP

iostat for the quick answer, biosnoop for the expert flourish.

POSSIBLE FOLLOW-UP

What if the writer is a kernel flusher thread — what does that imply?

Q243. How do you trace which config file a binary actually reads?**ADVANCED****SHORT ANSWER**

strace -e trace=openat -f ./binary 2>&1 | grep -E '\.conf' reveals every open attempt and failures (ENOENT), showing search order and the file actually used; ltrace for library-level calls; lsof for already-running processes.

INTERVIEW TIP

This strace one-liner solves 'why is my config ignored' definitively.

POSSIBLE FOLLOW-UP

The binary opens a config you didn't expect — what next?

Q244. A service starts, then dies within seconds. How do you catch why?**ADVANCED****SHORT ANSWER**

journalctl -u svc -f while starting; check exit code/signal in systemctl status; run ExecStart manually in foreground as the service user (sudo -u appuser); look for missing env (systemd has clean env), permissions, port conflicts, and OOM/cgroup kills (memory.events).

INTERVIEW TIP

Reproduce as the service user with the service's environment — that's the trick.

POSSIBLE FOLLOW-UP

Status shows signal=KILL — who could be sending it?

Q245. How do you determine if slowness is the server or the network?**ADVANCED****SHORT ANSWER**

Compare local vs remote behavior: curl the app on localhost (fast local = network/path suspect), measure with curl -w timing breakdown (DNS, connect, TTFB), mtr between endpoints for loss/latency hops, and ss -ti for retransmissions on established flows.

INTERVIEW TIP

`curl -w '%{time_namelookup} %{time_connect} %{time_starttransfer}'` — timing splits localize the layer.

POSSIBLE FOLLOW-UP

TTFB is slow but connect is fast — where is the problem?

Q246. How do you find what changed on a server recently?**ADVANCED****SHORT ANSWER**

Package history (dnf history / apt log), configuration drift (git-managed configs, etckeeper, Ansible runs), recent file changes: `find /etc -mtime -2`, auditd records, shell histories with timestamps, journal for restarts, and login records (last).

INTERVIEW TIP

`find /etc -mtime -2` is the fast, memorable forensic move.

POSSIBLE FOLLOW-UP

How would you make 'what changed' answerable by design?

Q247. df -h says 80% but the disk fills within minutes repeatedly. How do you catch the writer?**ADVANCED****SHORT ANSWER**

Watch growth live: watch df -h plus du snapshots to localize the directory; iotop/pidstat -d for the writing process; fatrace or eBPF opensnoop shows file opens in real time — identify the runaway log/dump generator and stop or truncate it.

INTERVIEW TIP

`fatrace` is the little-known gem for 'who is writing files right now'.

POSSIBLE FOLLOW-UP

How do you cap a service's disk writes going forward?

Q248. How do you debug intermittent 'connection reset' errors to a service?**ADVANCED****SHORT ANSWER**

Distinguish reset sources: app crash/restart (journal), idle timeout at LB/firewall, conntrack table full (dmesg), keepalive mismatches, or backlog overflow. Capture with tcpdump on both sides to see who sends RST and when relative to the request.

INTERVIEW TIP

The pcap showing who sends the RST turns argument into fact.

POSSIBLE FOLLOW-UP

The RST comes from a middlebox IP — what are your options?

Q249. A filesystem shows 'structure needs cleaning'. What do you do?**ADVANCED****SHORT ANSWER**

That's ext4 corruption: unmount (or rescue boot), run `e2fsck -f` on the device, review repairs; for XFS use `xfs_repair` (with `-L` only as documented last resort). Check `dmesg`/SMART for the root cause and validate data/backups after repair.

INTERVIEW TIP

Never fsck a mounted filesystem — say it before anything else.

POSSIBLE FOLLOW-UP

When is `xfs_repair -L` justified and what does it sacrifice?

Q250. Users can SSH but login takes 30 seconds. Why?**ADVANCED****SHORT ANSWER**

Classic causes: reverse DNS timeout on the server (UseDNS), GSSAPI authentication attempts, a slow LDAP/SSSD lookup, or `/etc/nsswitch` hitting a dead source. `ssh -vvv` timing shows the stalled phase; fix the resolver/SSSD or disable the offending option.

INTERVIEW TIP

'UseDNS/GSSAPI delay' is the textbook answer — name both immediately.

POSSIBLE FOLLOW-UP

How do you see which login phase stalls with `ssh -vvv`?

Q251. How do you verify whether a port block is on the host or in the network path?**ADVANCED****SHORT ANSWER**

On the server: `tcpdump port N` — if SYNs never arrive, the block is upstream (network FW/SG); if SYNs arrive with no reply, it's host firewall or no listener (check `ss -tlnp` and `iptables/nft` rules). Test outbound with `nc` from both ends.

INTERVIEW TIP

'Do SYNs reach the NIC?' is the single question that splits the problem.

POSSIBLE FOLLOW-UP

SYNs arrive and the host replies RST — what does that mean?

Q252. Log shows the app can't resolve a hostname only sometimes. How do you debug flapping DNS?**ADVANCED****SHORT ANSWER**

Query all configured servers individually (`dig @each`), check `resolved/dnsmasq` logs and cache stats, look for one bad upstream in rotation, UDP drops (firewall/conntrack), and low TTL churn; run a periodic `dig` loop logging failures with timestamps to correlate.

INTERVIEW TIP

Intermittent = one bad server in the pool until proven otherwise.

POSSIBLE FOLLOW-UP

How do `rotate`/`timeout` options in `resolv.conf` affect this?

Q253. How do you handle a runaway process consuming CPU that you cannot restart during business hours?**ADVANCED****SHORT ANSWER**

Contain it: renice +19, or better cgroup limit it live (systemctl set-property app.service CPUQuota=50%); if I/O too, IOWeight; capture perf data for later; schedule the real fix/restart in the window; document the constraint.

INTERVIEW TIP

systemctl set-property for live resource capping is the modern move.

POSSIBLE FOLLOW-UP

How does CPUQuota behave across multiple cores?

Q254. The same binary works on server A and fails on server B. Approach?**ADVANCED****SHORT ANSWER**

Diff the environments: OS/kernel versions, ldd binary for library resolution, env vars, ulimits, SELinux mode/contexts, filesystem mount options (noexec), CPU flags if optimized builds, and configs; strace on B to see the first failing syscall.

INTERVIEW TIP

'ldd + strace + env diff' is a compact, complete methodology.

POSSIBLE FOLLOW-UP

ldd shows 'not found' for one lib on B — what are the fixes?

Q255. How do you investigate high context switching?**ADVANCED****SHORT ANSWER**

vmstat's cs column and pidstat -w per process; distinguish voluntary (waiting on I/O/locks) vs involuntary (CPU contention). Causes: too many threads, lock contention, undersized CPU; fix via thread-pool sizing, affinity, or scaling.

INTERVIEW TIP

Voluntary vs involuntary switch distinction is the depth marker.

POSSIBLE FOLLOW-UP

What does high involuntary switching with a full run queue imply?

Q256. A backup job saturates the disk and hurts the app at 2 AM. Fix the contention.**ADVANCED****SHORT ANSWER**

Throttle the backup: ionice -c3 (idle class) or cgroup IO limits (IOReadBandwidthMax), rsync --bwlimit, run against an LVM snapshot to shorten app impact, and reschedule to the quietest window based on sar history.

INTERVIEW TIP

ionice idle-class for backups is the one-line classic answer.

POSSIBLE FOLLOW-UP

Why is backing up from a snapshot doubly beneficial?

Q257. How do you confirm packet loss and where it occurs?**ADVANCED****SHORT ANSWER**

mtr -rw target shows per-hop loss/latency over time (interpret intermediate-hop loss carefully — only endpoint loss matters definitively); ip -s link and ethtool -S for local drops; on servers, nstat/ss -ti for retransmits; test both directions since paths can be asymmetric.

INTERVIEW TIP

'Intermediate hop loss can be ICMP deprioritization' — that caveat shows mastery.

POSSIBLE FOLLOW-UP

Why can a middle hop show 20% loss while the end shows 0%?

Q258. What's your method when 'it worked yesterday' and nothing changed (allegedly)?**ADVANCED****SHORT ANSWER**

Trust telemetry, not memory: package/config change history, journal restarts, cert/token expiries (very common), cron/timer runs, disk/inode thresholds crossed, DNS/upstream dependency changes, and date-based triggers. Something changed — often outside the box (expiry, dependency, data growth).

INTERVIEW TIP

Name 'certificate expiry' early — it's the #1 'nothing changed' culprit.

POSSIBLE FOLLOW-UP

Which classes of failures trigger purely by time passing?

Q259. How do you safely test a risky change on a production server?**ADVANCED****SHORT ANSWER**

Snapshot/backup first (LVM/VM snapshot), announce a change window, keep an active rollback session (e.g., second root shell, iptables timed revert: sleep 300 && rollback), apply behind a canary/subset, verify health checks, then commit or roll back.

INTERVIEW TIP

The 'timed auto-revert for firewall/SSH changes' trick prevents lockouts — cite it.

POSSIBLE FOLLOW-UP

How do you avoid locking yourself out when changing sshd or firewall rules?

Q260. Describe end-to-end how you'd troubleshoot 'website is slow' on a LEMP server.**ADVANCED****SHORT ANSWER**

Split layers: client (curl -w timings), network (mtr), TLS (handshake time), nginx (access log \$request_time/\$upstream_response_time), PHP-FPM (slowlog, pool saturation), DB (slow query log, locks), and system (USE checks). The two timing fields in nginx instantly split web vs app vs DB.

INTERVIEW TIP

\$request_time vs \$upstream_response_time comparison is the pivotal trick.

POSSIBLE FOLLOW-UP

Upstream time is high but DB is fast — what remains?

SECTION A

Rapid Fire Round

50 one-line answers for last-minute revision

#	Question	Answer
1	Command to show current directory?	<code>pwd</code>
2	Kernel version command?	<code>uname -r</code>
3	PID 1 on modern Linux?	<code>systemd</code>
4	File storing user accounts?	<code>/etc/passwd</code>
5	File storing password hashes?	<code>/etc/shadow</code>
6	Numeric permission for <code>rw-r-xr-x</code> ?	<code>755</code>
7	Command to change file owner?	<code>chown</code>
8	Signal number for SIGKILL?	<code>9</code>
9	Default SSH port?	<code>22</code>
10	Command to follow a log live?	<code>tail -f (or journalctl -f)</code>
11	View free memory?	<code>free -h</code>
12	View disk space by filesystem?	<code>df -h</code>
13	View directory size?	<code>du -sh</code>
14	Modern replacement for <code>netstat</code> ?	<code>ss</code>
15	Modern replacement for <code>ifconfig</code> ?	<code>ip</code>
16	List listening TCP ports with processes?	<code>ss -tlnp</code>
17	Which process uses port 80?	<code>lsof -i :80</code>
18	Check SELinux mode?	<code>getenforce</code>
19	Restore SELinux contexts?	<code>restorecon -Rv</code>
20	Reload <code>systemd</code> after unit edits?	<code>systemctl daemon-reload</code>
21	Show failed services?	<code>systemctl --failed</code>
22	Logs of one service?	<code>journalctl -u <service></code>
23	Logs of previous boot?	<code>journalctl -b -1</code>
24	Default target check?	<code>systemctl get-default</code>
25	Cron entry for every 5 minutes?	<code>*/5 * * * *</code>
26	Edit your crontab?	<code>crontab -e</code>
27	One-time scheduled job tool?	<code>at</code>
28	Create a gzip tar archive?	<code>tar -czvf file.tar.gz dir/</code>
29	Extract a tar.gz?	<code>tar -xzvf file.tar.gz</code>

#	Question	Answer
30	Find files over 100MB?	find / -size +100M
31	Count lines in a file?	wc -l
32	Print 1st column with awk?	awk '{print \$1}'
33	Replace text in-place with sed?	sed -i 's/old/new/g' file
34	Show inode usage?	df -i
35	Show block devices?	lsblk
36	Show filesystem UUIDs?	blkid
37	Extend LV plus filesystem in one command?	lvextend -r -L +10G /dev/vg/lv
38	Software RAID status file?	/proc/mdstat
39	Check listening NFS exports of a server?	showmount -e server
40	Persistent hostname change?	hostnamectl set-hostname name
41	Test DNS against a specific server?	dig @8.8.8.8 domain
42	Show routing table?	ip route
43	Firewalld: list active rules?	firewall-cmd --list-all
44	Kernel messages command?	dmesg (or journalctl -k)
45	Kernel parameter tool?	sysctl
46	Trace syscalls of a PID?	strace -p PID
47	Per-process I/O live view?	iotop
48	Historical performance data tool?	sar
49	Zombie process state letter?	Z
50	Sticky bit permission on /tmp?	1777

SECTION B

Linux MCQs

50 multiple-choice questions with correct answers and short explanations

MCQ 1. Which file contains encrypted user passwords?

A. /etc/passwd B. /etc/shadow C. /etc/group D. /etc/login.defs

Correct Answer: B. /etc/shadow

Why: /etc/shadow stores password hashes and aging info, readable only by root.

MCQ 2. What does chmod 644 file grant?

A. rw-r--r-- B. rwxr--r-- C. rw-rw-r-- D. r--r--r--

Correct Answer: A. rw-r--r--

Why: 6=rw for owner, 4=r for group, 4=r for others.

MCQ 3. PID of the init/systemd process?

A. 0 B. 1 C. 2 D. 100

Correct Answer: B. 1

Why: The first user-space process is PID 1; all processes descend from it.

MCQ 4. Which signal cannot be caught or ignored?

A. SIGTERM B. SIGHUP C. SIGKILL D. SIGINT

Correct Answer: C. SIGKILL

Why: SIGKILL (and SIGSTOP) are handled by the kernel directly; processes can't trap them.

MCQ 5. Which command shows the last 20 lines of a file?

A. head -20 file B. tail -n 20 file C. cat -20 file D. less +20 file

Correct Answer: B. tail -n 20 file

Why: tail -n 20 prints the final 20 lines; head prints the first lines.

MCQ 6. Which is the modern socket statistics tool?

A. netstat B. ss C. ifconfig D. route

Correct Answer: B. ss

Why: ss replaces the deprecated netstat and is faster on busy systems.

MCQ 7. What does df -i show?

A. Disk I/O B. Inode usage C. Mounted devices D. Directory sizes

Correct Answer: B. Inode usage

Why: df -i reports inode totals/used/free — key when files can't be created despite free space.

MCQ 8. The sticky bit on a directory means:

A. Files run as owner B. Files inherit group C. Only owners can delete their files D. Directory is immutable

Correct Answer: C. Only owners can delete their files

Why: Sticky bit (e.g., /tmp = 1777) restricts deletion to file owners/root.

MCQ 9. Which runlevel equals multi-user.target?

A. 1 B. 3 C. 5 D. 6

Correct Answer: B. 3

Why: multi-user.target maps to classic runlevel 3 (non-graphical multi-user).

MCQ 10. Which command reloads unit files after editing them?

A. systemctl refresh B. systemctl daemon-reload C. systemctl reload systemd D. systemctl rescan

Correct Answer: B. systemctl daemon-reload

Why: daemon-reload re-reads unit definitions; 'reload' targets a service's own config.

MCQ 11. What does 2>&1 do?

A. Redirects stdin to stdout B. Redirects stderr to stdout C. Redirects stdout to stderr D. Discards all output

Correct Answer: B. Redirects stderr to stdout

Why: It duplicates fd 2 (stderr) onto fd 1 (stdout), combining the streams.

MCQ 12. Which finds files modified in the last 24 hours?

A. find / -mtime -1 B. find / -mtime +1 C. find / -atime 1 D. find / -newer 24

Correct Answer: A. find / -mtime -1

Why: -mtime -1 means modified less than 1 day ago; +1 means more than.

MCQ 13. Hard links cannot:

A. Share an inode B. Cross filesystems C. Survive original deletion D. Point to a file

Correct Answer: B. Cross filesystems

Why: Hard links reference an inode number, which is only unique within one filesystem.

MCQ 14. Default nice value of a process?

A. -20 B. 0 C. 10 D. 19

Correct Answer: B. 0

Why: Processes start at niceness 0; range is -20 (highest priority) to 19 (lowest).

MCQ 15. Which tool shows historical CPU usage from earlier today?

A. top B. sar -u C. ps aux D. uptime

Correct Answer: B. sar -u

Why: sar reads sysstat's collected history; top/ps only show the present.

MCQ 16. What does mkfs.xfs do?

A. Mounts XFS B. Repairs XFS C. Creates an XFS filesystem D. Resizes XFS

Correct Answer: C. Creates an XFS filesystem

Why: mkfs. formats a device with a filesystem; xfs_repair fixes, xfs_growfs grows.*

MCQ 17. Which LVM object do filesystems sit on?

A. PV B. VG C. LV D. PE

Correct Answer: C. LV

Why: You format and mount Logical Volumes; PVs feed VGs, which contain LVs.

MCQ 18. XFS filesystems can be:

A. Grown and shrunk online B. Only shrunk C. Grown but not shrunk D. Neither grown nor shrunk

Correct Answer: C. Grown but not shrunk

Why: XFS supports online grow (xfs_growfs) but has no shrink capability.

MCQ 19. Which RAID level is pure mirroring?

A. RAID 0 B. RAID 1 C. RAID 5 D. RAID 6

Correct Answer: B. RAID 1

Why: RAID 1 mirrors data across disks; RAID 0 stripes with no redundancy.

MCQ 20. Which command lists block devices in a tree?

A. fdisk -t B. lsblk C. blkls D. du -b

Correct Answer: B. lsblk

Why: lsblk shows disks, partitions, and mount points hierarchically.

MCQ 21. The OOM killer decision is influenced by:

A. File size B. oom_score_adj C. Nice value only D. Login shell

Correct Answer: B. oom_score_adj

Why: oom_score derives from memory use, adjusted by oom_score_adj per process.

MCQ 22. What does umask 027 give a new file?

A. 640 B. 750 C. 644 D. 600

Correct Answer: A. 640

Why: Files: 666-027 = 640 (dirs would be 750).

MCQ 23. Which SSH file holds keys allowed to log in to an account?

A. known_hosts B. id_rsa C. authorized_keys D. ssh_config

Correct Answer: C. authorized_keys

Why: The server checks ~/.ssh/authorized_keys for matching public keys.

MCQ 24. scp is built on top of:

A. FTP B. SSH C. NFS D. rsyncd

Correct Answer: B. SSH

Why: scp transfers files over the SSH protocol on port 22.

MCQ 25. Which rsync flag previews without changing anything?

A. --safe B. --dry-run C. --test D. --preview

Correct Answer: B. --dry-run

Why: --dry-run (-n) lists actions without executing them.

MCQ 26. Cron field order is:

A. min hour dom month dow B. hour min dom month dow C. min hour month dom dow D. dow min hour dom month

Correct Answer: A. min hour dom month dow

Why: Minute, hour, day-of-month, month, day-of-week, then the command.

MCQ 27. @reboot in crontab means:

A. Reboot the server B. Run at every boot C. Run before shutdown D. Run daily

Correct Answer: B. Run at every boot

Why: @reboot triggers the job once each time the system starts.

MCQ 28. Which shows the default gateway?

A. ip addr B. ip route C. ip link D. ip neigh

Correct Answer: B. ip route

Why: ip route prints 'default via <gw>' along with the routing table.

MCQ 29. DNS primarily uses which port?

A. 25 B. 53 C. 80 D. 443

Correct Answer: B. 53

Why: DNS queries use port 53 (UDP mostly, TCP for large/zone transfers).

MCQ 30. firewall-cmd changes persist only if you use:

A. --now B. --save C. --permanent (then --reload) D. --write

Correct Answer: C. --permanent (then --reload)

Why: Runtime rules vanish on reload/reboot unless added with --permanent and reloaded.

MCQ 31. REJECT differs from DROP because REJECT:

A. Silently discards B. Sends an error back C. Accepts then logs D. Rate limits

Correct Answer: B. Sends an error back

Why: REJECT returns ICMP/RST so clients fail fast; DROP causes timeouts.

MCQ 32. SELinux file contexts are viewed with:

A. ls -s B. ls -Z C. ls -C D. getfacl

Correct Answer: B. ls -Z

Why: ls -Z displays SELinux labels; getfacl shows POSIX ACLs.

MCQ 33. setenforce 0 puts SELinux into:

A. Disabled B. Permissive C. Enforcing D. Unconfined

Correct Answer: B. Permissive

Why: Permissive logs denials without blocking; full disable needs a config change and reboot.

MCQ 34. Which detects why SELinux denied an action?

A. sestatus B. audit2why C. semanage ports D. chcon -v

Correct Answer: B. audit2why

Why: audit2why interprets AVC denials from the audit log into human-readable causes.

MCQ 35. Too many CLOSE_WAIT sockets indicate:

A. Kernel bug B. Local app not closing sockets C. Firewall drops D. Normal behavior

Correct Answer: B. Local app not closing sockets

Why: CLOSE_WAIT persists until the local application calls close() — an app-side leak.

MCQ 36. Which shows per-disk utilization and await?

A. vmstat B. iostat -x C. free -m D. mpstat

Correct Answer: B. iostat -x

Why: iostat -x reports %util, await, and throughput per block device.

MCQ 37. High 'wa' in top means CPU is:

A. Running user code B. Idle waiting on disk I/O C. Handling interrupts D. Swapping code pages

Correct Answer: B. Idle waiting on disk I/O

Why: iowait counts idle time while uncompleted disk I/O is pending.

MCQ 38. Load average includes processes in state:

A. R only B. R and D C. S and Z D. T only

Correct Answer: B. R and D

Why: Linux counts runnable (R) plus uninterruptible sleep (D) tasks.

MCQ 39. Which limits open files for a systemd service?

A. limits.conf B. LimitNOFILE= C. ulimit in .bashrc D. fs.inotify

Correct Answer: B. LimitNOFILE=

Why: systemd services ignore PAM limits.conf; set LimitNOFILE in the unit.

MCQ 40. journalctl --since "1 hour ago" -p err shows:

A. All logs B. Errors from the last hour C. Kernel logs only D. One service's logs

Correct Answer: B. Errors from the last hour

Why: -p err filters priority error and worse; --since bounds the time window.

MCQ 41. Which regenerates initramfs on RHEL?

A. update-grub B. dracut -f C. mkfs.initrd D. grub2-install

Correct Answer: B. dracut -f

Why: dracut rebuilds the initramfs (update-initramfs on Debian/Ubuntu).

MCQ 42. rd.break on the kernel line is used to:

A. Enable debugging B. Break before pivot to reset root password C. Skip fsck D. Boot old kernel

Correct Answer: B. Break before pivot to reset root password

Why: rd.break stops in initramfs, letting you remount /sysroot and reset root's password.

MCQ 43. Which command shows what package owns /etc/nginx/nginx.conf on RHEL?

A. rpm -qf B. rpm -ql C. dnf list D. rpm -qa

Correct Answer: A. rpm -qf

Why: rpm -qf <file> maps a file back to its owning package; -ql lists a package's files.

MCQ 44. apt's low-level backend tool is:

A. yum B. dpkg C. rpm D. snap

Correct Answer: B. dpkg

Why: apt front-ends dpkg on Debian-family systems, as dnf front-ends rpm.

MCQ 45. Which tar flag lists archive contents?

A. -x B. -c C. -t D. -z

Correct Answer: C. -t

Why: tar -tvf archive lists; -c creates, -x extracts, -z means gzip.

MCQ 46. What does 'ln -s target link' create?

A. Hard link B. Symbolic link C. Copy D. Alias in shell

Correct Answer: B. Symbolic link

Why: ln -s makes a symlink — a path pointer that can cross filesystems.

MCQ 47. Which /proc file shows memory details?

A. /proc/cpuinfo B. /proc/meminfo C. /proc/mounts D. /proc/loadavg

Correct Answer: B. /proc/meminfo

Why: /proc/meminfo details totals, cache, buffers, hugepages, and more.

MCQ 48. nohup protects a job from which signal?

A. SIGKILL B. SIGHUP C. SIGSTOP D. SIGSEGV

Correct Answer: B. SIGHUP

Why: nohup ignores the hangup signal sent when the controlling terminal closes.

MCQ 49. Which command changes a running process's priority?

A. nice B. renice C. chrt --reset D. top -p

Correct Answer: B. renice

Why: renice adjusts niceness of an existing PID; nice sets it at launch.

MCQ 50. A '+' at the end of ls -l permissions means:

A. SELinux label B. ACL present C. Immutable file D. Sticky bit

Correct Answer: B. ACL present

Why: The plus sign indicates extended ACL entries; view them with getfacl.

DevOps Interview Kit
Rahul Shendage
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SECTION C

Linux Commands Cheat Sheet

100+ production commands organized by category

File Management

Task	Command	Example
List all files with details	ls	ls -lah
Show current directory	pwd	pwd
Change directory	cd	cd /var/log
Copy preserving attributes	cp	cp -a src/ dest/
Move / rename	mv	mv old.conf new.conf
Delete recursively	rm	rm -rf /tmp/build
Create nested directories	mkdir	mkdir -p /data/app/logs
Create empty file / update timestamp	touch	touch app.log
View file paginated	less	less /var/log/messages
First / last lines	head / tail	tail -n 50 app.log
Follow a log live	tail -f	tail -F /var/log/nginx/error.log
Find files by name	find	find / -name 'app.conf'
Find large recent files	find	find /var -size +100M -mtime -1
Fast indexed file search	locate	locate sshd_config
Hard / soft link	ln	ln -s /data/app /app
File type detection	file	file /usr/bin/passwd
Show file with line numbers	cat -n	cat -n script.sh
Compare two files	diff	diff -u old.conf new.conf
Show hidden attributes	lsattr / chattr	chattr +i /etc/resolv.conf

User Management

Task	Command	Example
Add user with home	useradd	useradd -m -s /bin/bash john
Set / change password	passwd	passwd john
Modify user (add group)	usermod	usermod -aG wheel john
Delete user with home	userdel	userdel -r john
Add group	groupadd	groupadd devops
Show identity and groups	id	id john
Current logged-in users	who / w	w
Login history	last	last -n 20
Password aging	chage	chage -l john
Lock / unlock account	usermod	usermod -L john

Task	Command	Example
Switch user	<code>su</code>	<code>su - john</code>
Run as root (audited)	<code>sudo</code>	<code>sudo systemctl restart nginx</code>
Edit sudoers safely	<code>visudo</code>	<code>visudo -f /etc/sudoers.d/app</code>

Permissions

Task	Command	Example
Change permissions (numeric)	<code>chmod</code>	<code>chmod 640 secret.conf</code>
Change permissions (symbolic)	<code>chmod</code>	<code>chmod u+x deploy.sh</code>
Recursive ownership change	<code>chown</code>	<code>chown -R app:app /opt/app</code>
Change group only	<code>chgrp</code>	<code>chgrp devops shared/</code>
Default permission mask	<code>umask</code>	<code>umask 027</code>
View ACLs	<code>getfacl</code>	<code>getfacl /data/report</code>
Grant ACL to a user	<code>setfacl</code>	<code>setfacl -m u:john:rw report.txt</code>
Set SUID / SGID / sticky	<code>chmod</code>	<code>chmod 2775 /data/shared</code>
Find SUID binaries	<code>find</code>	<code>find / -perm -4000 -type f</code>
Audit path permissions	<code>namei</code>	<code>namei -l /var/www/html/index.php</code>

Networking

Task	Command	Example
Show IP addresses	<code>ip addr</code>	<code>ip a</code>
Interface up/down	<code>ip link</code>	<code>ip link set eth0 up</code>
Routing table / gateway	<code>ip route</code>	<code>ip route show</code>
Test reachability	<code>ping</code>	<code>ping -c 4 8.8.8.8</code>
Trace network path	<code>traceroute</code> / <code>mtr</code>	<code>mtr -rw google.com</code>
DNS lookup	<code>dig</code>	<code>dig +short app.example.com</code>
DNS via specific server	<code>dig</code>	<code>dig @1.1.1.1 example.com</code>
Listening sockets + process	<code>ss</code>	<code>ss -tulnp</code>
Test remote port	<code>nc</code>	<code>nc -zv db01 5432</code>
Capture packets	<code>tcpdump</code>	<code>tcpdump -i eth0 port 443 -nn</code>
HTTP request / API test	<code>curl</code>	<code>curl -sI https://site.com</code>
Download files	<code>wget</code>	<code>wget -c https://host/file.iso</code>
Interface statistics/drops	<code>ip -s link</code>	<code>ip -s link show eth0</code>
NIC driver / ring info	<code>ethtool</code>	<code>ethtool -S eth0</code>
ARP/neighbor table	<code>ip neigh</code>	<code>ip neigh show</code>

Monitoring

Task	Command	Example
Live process view	<code>top</code> / <code>htop</code>	<code>htop</code>
Process list snapshot	<code>ps</code>	<code>ps aux --sort=-%mem head</code>
Load average / uptime	<code>uptime</code>	<code>uptime</code>

Task	Command	Example
Memory overview	<code>free</code>	<code>free -h</code>
Virtual memory / CPU stats	<code>vmstat</code>	<code>vmstat 2 5</code>
Disk I/O per device	<code>iostat</code>	<code>iostat -xz 2</code>
Per-process I/O	<code>iotop</code>	<code>iotop -oPa</code>
Historical stats	<code>sar</code>	<code>sar -u -f /var/log/sa/sa15</code>
Per-process CPU/mem/IO trend	<code>pidstat</code>	<code>pidstat -urd 2</code>
Kernel ring buffer	<code>dmesg</code>	<code>dmesg -T tail</code>
Open files by process/port	<code>lsof</code>	<code>lsof -i :8080</code>
Trace system calls	<code>strace</code>	<code>strace -f -p 1234</code>
Watch a command repeat	<code>watch</code>	<code>watch -n2 df -h</code>

Storage

Task	Command	Example
Filesystem usage	<code>df</code>	<code>df -hT</code>
Inode usage	<code>df -i</code>	<code>df -i /</code>
Directory sizes	<code>du</code>	<code>du -sh /var/* sort -rh</code>
Block device tree	<code>lsblk</code>	<code>lsblk -f</code>
UUIDs and fs types	<code>blkid</code>	<code>blkid /dev/sdb1</code>
Partition disk (GPT)	<code>parted</code>	<code>parted /dev/sdb mklabel gpt</code>
Partition disk (interactive)	<code>fdisk</code>	<code>fdisk /dev/sdb</code>
Create filesystem	<code>mkfs</code>	<code>mkfs.xfs /dev/sdb1</code>
Mount / unmount	<code>mount</code>	<code>mount /dev/sdb1 /data</code>
Test all fstab entries	<code>mount -a</code>	<code>mount -a</code>
Filesystem check (ext4)	<code>fsck</code>	<code>e2fsck -f /dev/sdb1</code>
Grow XFS online	<code>xfs_growfs</code>	<code>xfs_growfs /data</code>
Swap create/enable	<code>mkswap / swapon</code>	<code>swapon /swapfile</code>
SMART disk health	<code>smartctl</code>	<code>smartctl -a /dev/sda</code>
RAID status	<code>mdadm</code>	<code>mdadm --detail /dev/md0</code>

LVM

Task	Command	Example
Create physical volume	<code>pvcreate</code>	<code>pvcreate /dev/sdc</code>
List PVs / VGs / LVs	<code>pvs vgs lvs</code>	<code>vgs</code>
Create volume group	<code>vgcreate</code>	<code>vgcreate datavg /dev/sdc</code>
Extend VG with a disk	<code>vgextend</code>	<code>vgextend datavg /dev/sdd</code>
Create logical volume	<code>lvcreate</code>	<code>lvcreate -L 50G -n applv datavg</code>
Extend LV + filesystem	<code>lvextend</code>	<code>lvextend -r -L +20G /dev/datavg/applv</code>
Snapshot an LV	<code>lvcreate -s</code>	<code>lvcreate -s -L 5G -n snap /dev/datavg/applv</code>
Move data off a PV	<code>pvmove</code>	<code>pvmove /dev/sdc</code>

Firewall

Task	Command	Example
List firewalld rules	<code>firewall-cmd</code>	<code>firewall-cmd --list-all</code>
Open a service permanently	<code>firewall-cmd</code>	<code>firewall-cmd --permanent --add-service=https</code>
Open a port permanently	<code>firewall-cmd</code>	<code>firewall-cmd --permanent --add-port=8080/tcp</code>
Apply permanent rules	<code>firewall-cmd</code>	<code>firewall-cmd --reload</code>
List iptables rules	<code>iptables</code>	<code>iptables -L -n -v</code>
Allow a port (iptables)	<code>iptables</code>	<code>iptables -A INPUT -p tcp --dport 22 -j ACCEPT</code>
List nftables ruleset	<code>nft</code>	<code>nft list ruleset</code>
SELinux mode check	<code>getenforce</code>	<code>getenforce</code>
Fix SELinux contexts	<code>restorecon</code>	<code>restorecon -Rv /var/www</code>

SSH & Transfer

Task	Command	Example
Connect with key	<code>ssh</code>	<code>ssh -i key.pem user@host</code>
Copy key to server	<code>ssh-copy-id</code>	<code>ssh-copy-id user@host</code>
Generate keypair	<code>ssh-keygen</code>	<code>ssh-keygen -t ed25519</code>
Copy file to server	<code>scp</code>	<code>scp app.war user@host:/opt/</code>
Sync directories	<code>rsync</code>	<code>rsync -avz --delete src/ user@host:/dest/</code>
SFTP session	<code>sftp</code>	<code>sftp user@host</code>
Local port forward	<code>ssh -L</code>	<code>ssh -L 5432:db:5432 user@bastion</code>
Jump host connect	<code>ssh -J</code>	<code>ssh -J bastion user@internal</code>
Test sshd config	<code>sshd -t</code>	<code>sshd -t</code>

Package Management

Task	Command	Example
Install package (RHEL)	<code>dnf</code>	<code>dnf install nginx</code>
Update all (RHEL)	<code>dnf</code>	<code>dnf update -y</code>
Which package owns a file	<code>rpm -qf</code>	<code>rpm -qf /usr/sbin/sshd</code>
List package files	<code>rpm -ql</code>	<code>rpm -ql nginx</code>
Transaction history / undo	<code>dnf history</code>	<code>dnf history undo 42</code>
Search providers of a file	<code>dnf provides</code>	<code>dnf provides */semanage</code>
Install package (Debian)	<code>apt</code>	<code>apt install nginx</code>
Refresh repo metadata (Debian)	<code>apt update</code>	<code>apt update</code>
Which package owns a file (Debian)	<code>dpkg -S</code>	<code>dpkg -S /bin/ls</code>
List installed (Debian)	<code>dpkg -l</code>	<code>dpkg -l grep nginx</code>

Process Management

Task	Command	Example
Kill by PID (graceful)	<code>kill</code>	<code>kill 1234</code>
Force kill	<code>kill -9</code>	<code>kill -9 1234</code>

Task	Command	Example
Kill by name	<code>pkill</code>	<code>pkill -f java</code>
Find PID by name	<code>pgrep</code>	<code>pgrep -f nginx</code>
Process tree	<code>pstree</code>	<code>pstree -p 1</code>
Start with low priority	<code>nice</code>	<code>nice -n 10 ./batch.sh</code>
Change running priority	<code>renice</code>	<code>renice -n 5 -p 1234</code>
Background / foreground	<code>bg fg jobs</code>	<code>fg %1</code>
Survive logout	<code>nohup</code>	<code>nohup ./job.sh &</code>
Service control	<code>systemctl</code>	<code>systemctl restart app</code>
Service logs	<code>journalctl</code>	<code>journalctl -u app -f</code>

Compression

Task	Command	Example
Create tar.gz	<code>tar</code>	<code>tar -czvf logs.tar.gz /var/log/app</code>
Extract tar.gz	<code>tar</code>	<code>tar -xzvf logs.tar.gz -C /tmp</code>
List archive contents	<code>tar -t</code>	<code>tar -tvf logs.tar.gz</code>
Gzip / gunzip a file	<code>gzip</code>	<code>gzip -k huge.log</code>
Read gzipped file	<code>zcat / zgrep</code>	<code>zgrep ERROR app.log.gz</code>
Zip / unzip	<code>zip</code>	<code>zip -r site.zip /var/www</code>
High-ratio compression	<code>xz</code>	<code>xz -9 dump.sql</code>

Performance & Tuning

Task	Command	Example
Kernel parameter read/set	<code>sysctl</code>	<code>sysctl vm.swappiness=10</code>
Persist kernel parameters	<code>sysctl.d</code>	<code>echo 'vm.swappiness=10' > /etc/sysctl.d/99-tune.conf</code>
Show limits of a process	<code>/proc</code>	<code>cat /proc/1234/limits</code>
Set session limits	<code>ulimit</code>	<code>ulimit -n 65535</code>
CPU profiling	<code>perf</code>	<code>perf top</code>
Boot time analysis	<code>systemd-analyze</code>	<code>systemd-analyze blame</code>
Live cap a service's CPU	<code>systemctl</code>	<code>systemctl set-property app.service CPUQuota=50%</code>
I/O priority for a job	<code>ionice</code>	<code>ionice -c3 tar -czf backup.tgz /data</code>
Time a command	<code>time</code>	<code>time ./script.sh</code>

Text Processing

Task	Command	Example
Search pattern in files	<code>grep</code>	<code>grep -rn 'ERROR' /var/log/app/</code>
Print specific columns	<code>awk</code>	<code>awk -F: '{print \$1}' /etc/passwd</code>
Replace text	<code>sed</code>	<code>sed -i 's/8080/9090/g' app.conf</code>
Cut fields	<code>cut</code>	<code>cut -d: -f1,7 /etc/passwd</code>
Sort numerically reverse	<code>sort</code>	<code>sort -nr sizes.txt</code>

Task	Command	Example
Count duplicates	uniq	sort ips.txt uniq -c sort -nr
Build commands from input	xargs	find . -name '*.tmp' xargs rm -f
Split output to screen+file	tee	make 2>&1 tee build.log
Word/line count	wc	grep -c ERROR app.log

DevOps Interview Kit
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SECTION D

Linux Interview Cheat Code

100-row quick revision table: question, answer, command, tip

Interview Question	One-Line Answer	Command	Interview Tip
Check OS version	Distro name and release	<code>cat /etc/os-release</code>	Works on every modern distro
Check kernel version	Running kernel release	<code>uname -r</code>	Pair with <code>uname -a</code> for full info
Show system uptime and load	Uptime plus 1/5/15 load	<code>uptime</code>	Compare load to CPU core count
Count CPU cores	Logical CPU count	<code>nproc</code>	<code>lscpu</code> gives full topology
Find running process	Running processes	<code>ps -ef</code>	Mention <code>pgrep</code> for exact process lookup
Top memory consumers	Sorted process list	<code>ps aux --sort=-%mem head</code>	Or use <code>htop</code> 's sort keys
Top CPU consumers	Sorted by CPU	<code>ps aux --sort=-%cpu head</code>	Cross-check with <code>pidstat</code> trend
Kill process gracefully	SIGTERM first	<code>kill PID</code>	Escalate to <code>-9</code> only if needed
Kill all by name	Name-based kill	<code>pkill -f appname</code>	<code>-f</code> matches full command line
Find zombie processes	Z-state tasks	<code>ps -eo pid,ppid,stat,cmd awk '\$3~/Z/'</code>	Fix or restart the parent
Check who is logged in	Sessions and activity	<code>w</code>	<code>last</code> shows login history
Check last reboots	Boot history	<code>last reboot</code>	<code>who -b</code> for the latest boot
Check failed logins	Auth failures	<code>lastb</code>	Also <code>/var/log/secure</code> or <code>auth.log</code>
List open ports	Listening sockets	<code>ss -tulpn</code>	<code>netstat</code> is deprecated — say <code>ss</code>
Who uses port 8080	Owning process of a port	<code>lsof -i :8080</code>	<code>ss -tlnp grep 8080</code> also works
Check service status	Unit state and recent logs	<code>systemctl status nginx</code>	Follow with <code>journalctl -u</code> for depth
Enable service at boot	Creates wants symlink	<code>systemctl enable --now nginx</code>	<code>--now</code> also starts it immediately
List failed services	All failed units	<code>systemctl --failed</code>	First command after any reboot
Service logs live	Follow unit logs	<code>journalctl -u app -f</code>	<code>-xe</code> adds context after failures
Errors in last hour	Time+priority filter	<code>journalctl -p err --since '1 hour ago'</code>	<code>-p</code> warning widens the net
Previous boot logs	Logs before restart	<code>journalctl -b -1</code>	Essential for crash forensics
Kernel messages	Ring buffer with time	<code>dmesg -T tail -50</code>	<code>journalctl -k</code> is equivalent
Check disk space	Per-filesystem usage	<code>df -hT</code>	<code>-T</code> adds filesystem type

Interview Question	One-Line Answer	Command	Interview Tip
Check inode usage	Inode exhaustion check	<code>df -i</code>	Full inodes mimic full disk
Biggest directories	Space hogs by dir	<code>du -xh --max-depth=1 / sort -rh</code>	-x stays on one filesystem
Find huge files	Files over 500MB	<code>find / -xdev -size +500M -exec ls -lh {} +</code>	Add -mtime -1 for recent ones
Deleted but open files	Space held by deleted files	<code>lsof +L1</code>	Explains df vs du mismatch
Truncate a busy log	Free space without deleting	<code>: > /var/log/big.log</code>	Never rm an open log
Check memory	RAM and swap summary	<code>free -h</code>	'available' is the number that matters
Check swap activity	Swap in/out live	<code>vmstat 2</code>	Nonzero si/so = memory pressure
Top swap consumers	Per-process swap	<code>grep VmSwap /proc/*/status 2>/dev/null sort -k2 -nr head</code>	smem also reports swap
Clear page cache (test only)	Drop caches	<code>echo 3 > /proc/sys/vm/drop_caches</code>	Only for benchmarking, not a fix
Check disk I/O load	Device utilization	<code>iostat -xz 2</code>	await + %util tell saturation
Which process does I/O	Per-process disk I/O	<code>iotop -oPa</code>	pidstat -d 2 as alternative
Historical CPU data	Yesterday's performance	<code>sar -u -f /var/log/sa/saDD</code>	sysstat must be enabled
Trace a hung process	Live syscall trace	<code>strace -p PID</code>	Look for the blocking syscall
Library dependencies	Shared libs of a binary	<code>ldd /usr/bin/app</code>	'not found' = missing library
Check open file limit	Per-process limits	<code>cat /proc/PID/limits</code>	systemd needs LimitNOFILE
Count open FDs	FD usage of a process	<code>ls /proc/PID/fd wc -l</code>	Compare against the limit
Test network reachability	ICMP check	<code>ping -c4 host</code>	Blocked ICMP ≠ host down
Test a TCP port	Port connectivity	<code>nc -zv host 443</code>	curl -v telnet://host:443 too
Trace network path	Per-hop loss/latency	<code>mtr -rw host</code>	Endpoint loss is what counts
DNS quick lookup	Resolve a name	<code>dig +short name</code>	@server isolates the resolver
Check resolver config	Active DNS servers	<code>resolvectl status</code>	Or cat /etc/resolv.conf
Show default gateway	Routing table	<code>ip route</code>	Missing default = no internet
Show interface errors	Drops and errors	<code>ip -s link show eth0</code>	ethtool -S for NIC counters
Capture traffic on a port	Packet capture	<code>tcpdump -i any port 443 -nn</code>	-w file.pcap for Wireshark
Check TCP retransmits	Per-connection TCP info	<code>ss -ti</code>	nstat for global counters
List firewall rules	Active firewalld config	<code>firewall-cmd --list-all</code>	iptables -L -n for raw view
Open port permanently	Persistent firewall change	<code>firewall-cmd --permanent --add-port=8080/tcp && firewall-cmd --reload</code>	Forgetting --reload is the classic miss
SELinux current mode	Enforcing or permissive	<code>getenforce</code>	sestatus for full detail

Interview Question	One-Line Answer	Command	Interview Tip
Recent SELinux denials	AVC denial search	<code>ausearch -m AVC -ts recent</code>	audit2why explains them
Fix web dir contexts	Restore default labels	<code>restorecon -Rv /var/www</code>	Fix contexts, don't disable SELinux
Add user to group	Supplementary group add	<code>usermod -aG docker john</code>	Without -a you wipe other groups
Check user identity	UID, GID, groups	<code>id john</code>	getent passwd tests NSS too
Lock a user account	Disable password login	<code>usermod -L john</code>	chage -E0 fully expires it
Force password change	Expire password now	<code>chage -d 0 john</code>	chage -l shows aging policy
Grant limited sudo	Command-specific sudo	<code>echo 'john ALL=(root) NOPASSWD:/usr/bin/systemctl restart app' visudo -f /etc/sudoers.d/john -</code>	Absolute paths, no wildcards
Find SUID binaries	Privilege-escalation audit	<code>find / -perm -4000 -type f 2>/dev/null</code>	Baseline and diff regularly
World-writable files	Permission audit	<code>find / -xdev -type f -perm -0002</code>	Review before mass chmod
File permissions along a path	Full path audit	<code>namei -l /var/www/html/app.php</code>	Finds the missing execute bit fast
Give one user ACL access	Extra permission without chown	<code>setfacl -m u:john:rx /data/reports</code>	getfacl verifies; + in ls -l flags it
Make file immutable	Protect from edits/deletes	<code>chattr +i /etc/critical.conf</code>	lsattr reveals it; even root blocked
Change hostname	Persistent hostname	<code>hostnamectl set-hostname web01</code>	Update /etc/hosts too
Sync time now	Force clock step	<code>chronyc makestep</code>	chronyc tracking shows drift
Check time sync status	NTP source health	<code>chronyc sources -v</code>	Kerberos breaks past ~5 min skew
Schedule job every 5 min	Cron schedule	<code>* /5 * * * * /opt/script.sh >> /var/log/job.log 2>&1</code>	Always capture output in cron
Edit root's crontab	Root scheduled jobs	<code>sudo crontab -e</code>	System jobs go in /etc/cron.d
One-time job tonight	at scheduling	<code>echo '/opt/task.sh' at 23:00</code>	atq lists, atrm removes
List systemd timers	Modern scheduled jobs	<code>systemctl list-timers</code>	Timers give logs and dependencies
Archive a directory	Compressed tarball	<code>tar -czvf app-\$(date +%F).tar.gz /opt/app</code>	Date-stamp your archives
Extract to a target dir	Controlled extraction	<code>tar -xzvf app.tar.gz -C /tmp/restore</code>	-t lists before extracting
Sync dirs between servers	Incremental transfer	<code>rsync -avz --delete /src/ host:/dest/</code>	--dry-run before --delete
Resume a big download	Continue partial file	<code>wget -c URL</code>	curl -C - does the same
Test an HTTPS endpoint	Headers and status	<code>curl -sIv https://api.example.com</code>	-w timings localize slowness
Check certificate expiry	TLS cert dates	<code>openssl s_client -connect host:443 -servername host 2>/dev/null openssl x509 -noout -dates</code>	Missing intermediates = common failure

Interview Question	One-Line Answer	Command	Interview Tip
Generate SSH key	Modern keypair	<code>ssh-keygen -t ed25519</code>	ed25519 over rsa today
Copy SSH key to host	Enable key login	<code>ssh-copy-id user@host</code>	Check ~/.ssh perms if it fails
SSH via bastion	Jump host in one line	<code>ssh -J bastion user@internal</code>	Cleaner than nested ssh
Tunnel a remote DB port	Local forward	<code>ssh -L 5432:db:5432 user@bastion</code>	Access DB at localhost:5432
Find package owning a file	Reverse file lookup	<code>rpm -qf /usr/sbin/nginx</code>	dpkg -S on Debian
Rollback last update	Undo transaction	<code>dnf history undo last</code>	Snapshot first when possible
List repo sources	Configured repositories	<code>dnf repolist</code>	Stale repos cause dependency errors
Services needing restart	Post-patch check	<code>dnf needs-restarting -s</code>	Patching completes at restart
Check block devices	Disk/partition tree	<code>lsblk -f</code>	Includes fs types and UUIDs
Get UUID for fstab	Stable device reference	<code>blkid /dev/sdb1</code>	Never use /dev/sdX in fstab
Validate fstab safely	Test before reboot	<code>mount -a</code>	Catches bad entries early
Extend LV and filesystem	Online growth	<code>lvextend -r -L +10G /dev/vg/lv</code>	-r resizes the FS automatically
Check RAID health	Array status	<code>cat /proc/mdstat</code>	mdadm --detail for full view
Check disk SMART health	Predict failures	<code>smartctl -a /dev/sda</code>	Reallocated sectors = replace soon
Find what's writing files now	Live file access trace	<code>fatrace</code>	Great for mystery disk growth
Show mounted NFS exports	Server's export list	<code>showmount -e nfsserver</code>	rpcinfo -p checks services
Release a busy mount	Find blockers then unmount	<code>fuser -vm /mnt && umount /mnt</code>	umount -l as last resort
Boot time breakdown	Slow boot analysis	<code>systemd-analyze blame</code>	critical-chain shows the path
Rebuild initramfs	After driver/storage change	<code>dracut -f</code>	update-initramfs -u on Debian
Set kernel parameter now	Runtime tuning	<code>sysctl -w vm.swappiness=10</code>	Persist in /etc/sysctl.d/
Cap service memory	cgroup memory limit	<code>systemctl set-property app.service MemoryMax=2G</code>	Also edit the unit for permanence
Watch anything repeatedly	Auto-refresh command	<code>watch -n2 'ss -s'</code>	Great for live incident views
Record terminal session	Evidence of change work	<code>script /tmp/change-\$(date +%F).log</code>	Auditors love this
Find text across configs	Recursive grep	<code>grep -rn 'listen 80' /etc/nginx/</code>	-l lists only filenames
Sum a log column	awk aggregation	<code>awk '{sum+=\$10} END{print sum}' access.log</code>	Adapt \$10 to your format
Top IPs in access log	Frequency report	<code>awk '{print \$1}' access.log sort uniq -c sort -nr head</code>	The classic pipeline — memorize it

You are interview-ready.

Revise the Cheat Code table the night before, the Rapid Fire round one hour before, and walk in with the 60-second triage checklist in your head.

Linux Interview Kit — 2026 Edition — Rahul Shendage

DevOps Interview Kit
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